

Civic.ai - A Citizen Centric AI-Powered Government Service Assistant

by

Md Shihab Sarker

22101516

G.Araf Ahmed

22101532

Tapu Mohammad Fahad

24241263

Fahim Foysal Abir

22101543

Mohammad Zunaid

24241264

A thesis submitted to the Department of Computer Science and Engineering in partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
Brac University

Final thesis working manuscript

Evidence snapshot: 24 September 2026

Submission semester and approval details: awaiting author confirmation

© 2026. Brac University. All rights reserved.

Declaration

This manuscript is prepared for the authors' review and signature. The authors must confirm that the work submitted is their own, that external sources are appropriately acknowledged and that the final document complies with the university's academic-integrity and tool-disclosure requirements.

Author confirmation and signatures

Md Shihab Sarker (22101516) _____

G.Araf Ahmed (22101532) _____

Tapu Mohammad Fahad (24241263) _____

Fahim Foysal Abir (22101543) _____

Mohammad Zunaid (24241264) _____

Approval

Administrative page pending completion by the authors and examining committee. No approval or acceptance is asserted in this working manuscript.

Thesis: Civic.ai - A Citizen Centric AI-Powered Government Service Assistant.

Thesis ID: T2520877. Submission semester and examination date require confirmation; the previous submission contains inconsistent dates.

Supervisor	Name, designation and signature to be confirmed
------------	---

Co-supervisor, if applicable	Name, designation and signature to be confirmed
------------------------------	---

Thesis coordinator	Name, designation and signature to be confirmed
--------------------	---

Head of department	Name, designation and signature to be confirmed
--------------------	---

Ethics Statement

This project develops an informational assistant from collected government-service documents and prepared text derivatives. It does not issue official decisions, authenticate identities, process payments or approve applications. Public availability is not treated as an automatic redistribution licence. Original documents, derived text and source links must retain appropriate provenance and usage restrictions.

The manuscript includes an aggregate analysis of the team’s needs-analysis questionnaire. The analysis excludes explicit non-consent and restricts substantive reporting to respondents who declared an adult age group; responses from minors and those with missing age are held out pending clarification of consent arrangements. Raw records and free-text responses are not included in the public thesis bundle. No ethics-board approval, representative population survey, completed chatbot usability study or independently adjudicated expert benchmark is claimed. Informal developer and domain-informed observations remain distinct from systematic human evaluation. Personal identification numbers, scans and private application records are not needed for the reported fixed-question experiments and should not be collected by a public deployment without appropriate safeguards.

The project used AI-assisted coding, debugging and manuscript preparation. The authors remain responsible for reviewing the implementation, references, analysis and final submission. Model-generated answers and automatic judges are not authoritative interpretations of government rules. Review flags and incomplete outputs are disclosed rather than represented as verified advice.

Abstract

Accessing government services in Bangladesh is often difficult due to confusing procedures, lack of clear and centralized information and reliance on middlemen. This prevents many citizens from accessing these services efficiently, resulting in unnecessary time delays and financial burdens. This thesis aims to solve this problem by developing an AI-Powered interactive system that provides citizen centric guidance for certain government services which will act as a digital bridge between citizens and government services. This system will inform users about required procedures and credentials of the respective services by offering dialogue and question-answering sessions. The outcome is a user-friendly, accessible system that empowers citizens with correct information, reduces dependency on intermediaries and simplifies interactions with government offices.

Keywords: Artificial Intelligence (AI), Government Services, Intermediaries, AI-Powered interactive system, Citizen centric guidance, Digital bridge, Dialogue and question-answering sessions, User-friendly and accessible system.

Dedication

The authors' dedication text is awaiting confirmation. This page is retained in the supplied template's position.

Acknowledgment

We acknowledge the guidance provided during the thesis process and the public institutions whose service documents informed the knowledge corpus. We also acknowledge the developers of the open-source retrieval, embedding and local language-model software used in the implementation. Individual acknowledgements will be confirmed by the authors before submission.

Table of Contents

Declaration	i
Approval	ii
Ethics Statement	iii
Abstract	iv
Dedication	v
Acknowledgment	vi
Table of Contents	vii
List of Figures	x
List of Tables	xi
Nomenclature	xiii
1 Introduction	1
1.1 Background	1
1.2 Rational of the Study or Motivation	2
1.3 Problem Statement	2
1.4 Objective	3
1.5 Methodology in Brief	3
1.6 Scopes and Challenges	4

1.7	Team Overview	5
1.8	Key Learnings and Insights	5
2	Literature Review	6
2.1	Preliminaries	6
2.2	Review of Existing Research	7
2.3	Summary of Key Findings	10
3	Requirements, Impacts and Constraints	11
3.1	Final Specifications and Requirements	11
3.2	Societal Impact	12
3.3	Environmental Impact	13
3.4	Ethical Issues	13
3.5	Standards - if applicable	14
3.6	Project Management Plan	14
3.7	Risk Management	15
3.8	Economic Analysis	16
4	Proposed Methodology	17
4.1	Design Process or Methodology Overview	17
4.2	Preliminary Design or Design (Model) Specification	17
4.3	Data Collection -(If Applicable)	20
4.3.1	Data Cleaning	22
4.3.2	Data Transformation	23
4.3.3	Data Integration	23
4.3.4	Data Reduction	24
4.3.5	Summary of Preprocessed Data	25
4.4	Implementation of Selected Design	26
5	Result Analysis	32
5.1	Performance Evaluation	32

5.2	Analysis of Design Solutions	38
5.3	Final Design Adjustments	40
5.4	Statistical Analysis	40
5.5	Comparisons and Relationships	42
5.6	Discussions	43
6	Conclusion	44
6.1	Summary of Findings	44
6.2	Contributions to the Field	45
6.3	Recommendations for Future Work	45
	Bibliography	47
	Appendix A How to install L^AT_EX	49
	Appendix B Overleaf: GitHub for L^AT_EX projects	50

List of Figures

4.1	Complete CivicRAG architecture	18
4.2	Comparison and evaluation boundaries	19
4.3	Source-preserving preparation of the active corpus: original materials remain available while normalized derivatives are organized into identifiable retrieval units. Separate cleanup candidates are not substituted for the evaluated corpus.	22
4.4	How answer evidence remains connected to a vector without being generated by the database. BM25 is a separate index of r_i , omitted here to isolate the dense-storage link. All arrows entering storage denote stored fields, not additional embeddings.	24
4.5	Active corpus composition from the frozen evidence manifest: 3,709 chunks and 174 prepared source units. Chunk share reflects document length and splitting, not demand, accuracy or unique facts.	25
5.1	Stated willingness and conditional trust in the needs-analysis questionnaire. Only consenting adult submissions are included; denominators differ because of missing answers.	33
5.2	Requested assistance, based on explicit selections among 77 valid responses. Multiple choices were allowed. One “all options” response is not expanded, so explicit-choice counts are conservative. Requested features are not necessarily implemented capabilities.	34
5.3	Completed local self-judge scores for matched saved answers, with 30 valid scores per system and measure. Custom context relevance is binary and distinct from standard context precision/recall.	36

List of Tables

2.1	Architectural relationship to NextRAG; the systems address different datasets and tasks.	8
3.1	Requirement-to-implementation mapping for the final system.	12
3.2	Completed development milestones documented by repository revisions. .	15
3.3	Implementation-specific risks, controls and residual limitations.	15
4.1	Design decisions and implementation-specific justifications.	20
4.2	Active corpus inventory computed from configured files. Source units are prepared documents, not independent facts.	25
4.3	Selected unresolved active-corpus flags, counted once per derivative document. These counts may overlap.	26
4.4	Controlled and varied elements of the main comparison.	31
5.1	Needs-analysis responses among consenting adults. Denominators exclude blank answers to each item.	32
5.2	Operational measurements for the matched 30-question runs. Times are seconds; runs occurred separately.	35
5.3	Completed saved-answer evaluation: Qwen3 self-judge scores, not independent accuracy.	36
5.4	Overall: provisional retrieval-only scores with assistant-reviewed labels. Hit, precision and MRR use 26 questions; pooled recall and nDCG use 25. .	38
5.5	Paired CivicRAG minus Simple RAG differences with descriptive bootstrap intervals.	41
5.6	Passport: provisional retrieval-only scores with assistant-reviewed labels. Hit, precision and MRR use 8 questions; pooled recall and nDCG use 8. .	41

5.7	Birth registration: provisional retrieval-only scores with assistant-reviewed labels. Hit, precision and MRR use 9 questions; pooled recall and nDCG use 9.	41
5.8	BRTA: provisional retrieval-only scores with assistant-reviewed labels. Hit, precision and MRR use 9 questions; pooled recall and nDCG use 8. .	42
5.9	Secondary generator runs on the selected-evidence configuration. Language rejections are operational events, not factual error counts.	42

Nomenclature

BDRIS	Birth and Death Registration Information System
BRTA	Bangladesh Road Transport Authority
BM25	Best Matching 25 lexical ranking method
LLM	Large language model
MRP	Machine-readable passport
NFC	Unicode Normalization Form C
OCR	Optical character recognition
QA	Question answering
RAG	Retrieval-augmented generation
RRF	Reciprocal rank fusion

Chapter 1

Introduction

1.1 Background

Obtaining information about a government service often involves more than finding a website. A citizen must identify the correct procedure, understand the required documents and determine which conditions apply to their situation. In the Bangladeshi materials collected for this project, information is distributed across frequently asked questions, citizen charters, application instructions, fee tables and legal documents. Birth registration, passport applications and driving licences illustrate this difficulty: similar words can refer to different stages of a service.

This creates a gap between how information is published and how citizens ask for it. A document may organize requirements by legal provision, while a user asks a short question in everyday Bangla. Keyword search helps locate exact terms, but wording differences can obscure a relevant passage. Conversely, a result containing the right service name may describe the wrong procedure. Instructions for collecting a passport do not necessarily answer a question about applying for one.

Large language models (LLMs) offer a conversational way to explain information. However, an answer generated from model parameters alone need not reflect the supplied government documents or preserve their conditions. Retrieval-Augmented Generation (RAG) addresses this problem by supplying external passages during generation [1]. Source-grounded retrieval makes the evidence inspectable and permits knowledge updates separately from model training. It improves the basis for answering, without guaranteeing that every retrieved passage or generated statement is correct.

In this context, we developed *Civic.ai – A Citizen-Centric AI-Powered Government Service Assistant*. Its implemented pipeline, referred to as CivicRAG, connects a predominantly Bangla civic-service corpus to a local question-answering interface. The current

knowledge base covers birth registration, passport services and Bangladesh Road Transport Authority (BRTA) services. Civic.ai provides document-based guidance; it does not submit applications, access personal government records or replace the responsible authority.

1.2 Rational of the Study or Motivation

Our motivation combines an information-access need with a data-engineering problem. Citizens need understandable explanations, whereas published service information must first be collected and organized before a retrieval system can use it. Preparing this knowledge resource is a central contribution of the project.

The team's needs-analysis questionnaire supports this direction. Among valid responses from consenting adults, 55 of 77 submissions (71.4%) indicated willingness to use the described AI assistant instead of an intermediary. Required-document guidance was selected by 62 of 77 (80.5%), and payment or appointment information by 58 of 77 (75.3%). These findings motivated our emphasis on requirements, procedures and source-traceable guidance. They describe interest in a prospective service, not measured adoption of Civic.ai.

Trust also influenced the design. Even under the questionnaire's hypothetical verified-information condition, 42 of 76 respondents (55.3%) would recheck an answer. Therefore, the interface exposes evidence and available source links rather than asking users to rely on a response alone. Chapter 4 explains the survey processing and item-specific denominators; the detailed findings belong to the results chapter.

We selected local inference to support experimentation without sending each question and its evidence to a hosted generation service. This creates a trade-off between accessible deployment and response time. We consequently emphasize a manageable three-domain scope, reusable evaluation records and clear evidence boundaries rather than attempting every government service at once.

1.3 Problem Statement

The problem addressed by this thesis is how to turn heterogeneous civic-service documents into a usable, source-traceable Bangla question-answering system. The difficulty begins with the data: headings, conditions, tables and provenance can be lost during conversion or chunking. If these relationships are broken, later retrieval cannot reliably recover the intended meaning.

The next difficulty is applicability. A useful result must concern both the service and the

action requested by the user. Finally, the generator must express that evidence correctly: it may otherwise omit a condition, combine unrelated requirements or return an incomplete answer. These stages are connected, but require separate examination.

We therefore investigate three questions:

1. How can collected materials become reusable domain-specific evidence units while preserving source relationships and unresolved quality issues?
2. How does CivicRAG’s hybrid retrieval and evidence-selection configuration compare with a dense-only Simple RAG baseline under matched generation conditions?
3. How can retrieval behaviour and answer quality be evaluated separately to identify the strengths and limitations of the implemented system?

1.4 Objective

Our **research objectives** are to develop and document the three-domain civic-service dataset, examine the selected RAG configuration against a matched baseline, and analyse retrieval and generation using reproducible evidence. Dataset preparation and representation are described in Chapter 4; comparative observations and evaluation limitations are addressed in Chapter 5.

Our **system objectives** are to support Bangla questions within a selected service domain, retrieve evidence through semantic and lexical matching, generate a Bangla explanation from the supplied material, and retain the passages and identifiers needed to inspect a response. Additional engineering objectives are local execution, configurable models and reversible changes through version control. These objectives inform the requirements in Chapter 3.

The study adapts established RAG techniques rather than introducing a new foundation model. Indexing documents is knowledge-base construction, not model training. The intended contribution is a developed and evaluated civic-service system supported by a traceable dataset, with claims limited to implemented functions and recorded experiments.

1.5 Methodology in Brief

The methodology separates offline preparation from online answering. Offline, original materials are preserved, prepared text is normalized, and document-aware chunks receive identifiers and metadata. Each chunk retains searchable text and source evidence. BGE-M3 embeds the searchable representation [2]; ChromaDB stores vectors and linked text. A separate BM25 index supports lexical retrieval.

Online, the question is normalized and searched within the selected domain. Dense and BM25 rankings are combined using weighted Reciprocal Rank Fusion (RRF) [3], followed by reranking and automatic applicability-based evidence selection. Selected passages are passed to a local generator through Ollama. The main paired comparison uses Qwen3:8b [4], with checks for unsuitable-language output and incomplete generation.

The modular pipeline permits independent inspection of preparation, retrieval and generation. Saved questions, contexts, answers and configuration records support evaluation without repeatedly generating answers. A matched dense-only baseline provides the system comparison, while retrieval-only experiments examine ranked passages separately.

1.6 Scopes and Challenges

Current scope. Civic.ai handles single-turn informational questions using three domain-specific collections. Sources contain Bangla, English and mixed-language content, but the principal interface and comparison target Bangla questions and Bangla responses. The 30-question comparison contains ten passport, ten birth-registration and ten driving-licence questions. The retained registration source collection includes combined birth-and-death legal material, but the implemented study and its evaluated tasks focus on birth registration. The storage identifier is retained for reproducibility; it is not a claim of evaluated death-registration capability.

Data challenges. Uneven formatting, scanned text, missing URLs and historical instructions complicate preparation. We retain provenance and review flags rather than treating every converted document as independently verified.

Retrieval challenges. Different procedures share vocabulary, and broad questions can require several sections. Domain selection reduces cross-service interference, but relevance and coverage still require evaluation.

Generation challenges. Local models may omit conditions, combine incompatible evidence or use unnatural Bangla. Output limits can interrupt long checklists. Direct, paraphrased and scenario-based questions use the same pipeline, although the main set does not establish robustness across all these variants.

Future expansion. Additional domains, more varied unseen questions and stronger hardware are possible extensions. Transaction execution, document authentication and live application-status lookup remain outside the implementation. Reused development questions support a bounded comparison, not unrestricted performance claims.

1.7 Team Overview

The project team consists of Md Shihab Sarker, G.Araf Ahmed, Tapu Mohammad Fahad, Fahim Foysal Abir and Mohammad Zunaid. Our work brings together source collection, machine-readable document preparation, pipeline development, experimental review and thesis documentation. These activities are interdependent: preparation changes can affect retrieval, while answer failures can expose missing conditions in evidence.

Source directories, experiment records and Git history keep the outputs traceable. They are used to document the project rather than infer individual responsibilities or hours worked from filenames. Section 3.6 describes the completed development sequence and its controls.

1.8 Key Learnings and Insights

The principal architectural lesson is that finding a related document and answering the question are different tasks. Combining retrieval methods creates opportunities to find evidence, but additional passages can introduce distractions. Selection therefore needs to preserve service conditions rather than simply increase the amount of context.

Dataset development demonstrated the value of original-file preservation, headings and stable identifiers. Bangla processing showed that encoding consistency is necessary but insufficient: normalized text may still contain OCR errors or ambiguous wording. Both structure and language quality affect downstream behaviour.

Finally, evaluation must distinguish retrieval quality, answer support and usefulness. Source IDs make an answer inspectable, but do not prove each statement. A longer response may cover more cases without being more precise. These lessons shape the requirements and methodology in the following chapters.

Chapter 2

Literature Review

2.1 Preliminaries

This section introduces the concepts needed to understand our design. Preparing knowledge, finding applicable passages and generating an explanation are distinct activities; they should not be treated as one model operation.

Retrieval-Augmented Generation. RAG supplies external evidence to a language model at inference time [1]. A simplified formulation is $a = G(q, E_q)$, where q is the question, E_q is the evidence and a is the response. This makes the supporting information available separately from the model parameters. In Civic.ai, document updates do not require retraining the answer model, and the supplied context is retained with each saved result.

Document Understanding and Structured Text. Document conversion must preserve more than words: headings, tables and reading order help determine which conditions govern a statement. Docling is a document-understanding framework for structured PDF conversion, including layout analysis and table recognition, with machine-readable outputs such as Markdown [5]. In Civic.ai, the confirmed ingestion stage reads team-prepared Markdown and structured JSON; the archived records do not establish Docling as the converter used for those derivatives. We distinguish this relevant preprocessing framework from the verified implemented stages.

Dense Retrieval and Vector Storage. Dense retrieval encodes queries and passages into vectors and ranks them by proximity. BGE-M3 supports multilingual representations, motivating its use with our Bangla and English source material [2]. ChromaDB associates

embeddings, document text and metadata through shared identifiers. In Civic.ai, BGE-M3 supplies dense vectors and a persistent Chroma collection per domain links results back to evidence. The storage choice supports local operation; ChromaDB does not generate answers.

Lexical Retrieval and Rank Fusion. BM25 uses term frequency, inverse document frequency and length normalization to rank lexical matches [6]. It can retain exact service terms and form labels when those words matter. RRF merges ranked lists without assuming their raw scores share a common scale [3]. In Civic.ai, a separate BM25 index complements dense search, and weighted RRF combines the rankings. Hybrid retrieval means combining these channels; RRF is our chosen combination method, not the only possible one.

Reranking and Evidence Selection. A cross-encoder evaluates a query and passage together, allowing finer interaction after initial retrieval [7]. This citation supports the reranking approach, not the identity of our configured checkpoint. The multilingual BGE reranker is configured for this stage. Evidence selection then determines which candidates fit the service and action within a finite prompt budget. In Civic.ai, the selector is a heuristic function, not a second LLM or a human choosing passages during normal operation. Its trace makes decisions inspectable, but does not certify relevance.

Local Generation and Evaluation. Ollama provides a local model-serving interface. Qwen3 supports thinking and non-thinking operation [4]; the principal experiment uses the latter. RAGAS supplies model-assisted measures of RAG outputs [8]. In Civic.ai, generation and saved-answer evaluation are separate processes: evaluating an existing response does not require generating it again. Answer support and relevance are also distinguished from passage-ranking quality.

2.2 Review of Existing Research

Existing work addresses complementary parts of the problem. General RAG provides the retrieval-generation foundation, multilingual systems address language coverage, and Bangla-focused research examines context use and verification. These contributions inform our design but do not remove the need to collect and structure applicable civic-service evidence.

Bilingual Retrieval and NextRAG. NextRAG is our principal architectural inspiration. It studies bilingual financial question answering using dense retrieval, BM25 and Dirichlet-smoothed query-likelihood retrieval, followed by weighted RRF, reranking and language-aware generation [9]. Its relevance is methodological: heterogeneous Bangla and English documents motivate complementary retrieval channels. Financial questions are not treated as a government-service benchmark or as directly comparable results.

Compared with NextRAG’s three retrieval channels, CivicRAG uses two: BGE-M3 dense retrieval and BM25. We retain fusion and reranking while introducing our civic corpus, linked search/evidence representations and applicability selection. This is an adaptation rather than a replication of the paper’s data, models or reported scores.

Table 2.1 summarizes this relationship without implying a cross-dataset performance comparison. The shared components explain our starting point. The different evidence re-

Table 2.1: Architectural relationship to NextRAG; the systems address different datasets and tasks.

Aspect	NextRAG	CivicRAG
Knowledge domain	Financial documents	Birth registration, passport and BRTA documents
Retrieval	Dense, BM25, Dirichlet QLM	Dense and BM25
Combination	Weighted RRF and reranking	Weighted RRF and reranking
Context handling	Bilingual retrieval/generation design	Applicability selection and whole-chunk budget
Evaluation setting	Financial question answering	Matched civic questions and separate retrieval tests

quirements explain why our measured comparison uses Simple RAG on the same civic corpus rather than the paper’s numerical results.

Bangla Context Use and Verification. TraSe investigates Bangla question answering through translative prompting and answer selection, including automatic and human-in-the-loop retrieval [10]. In contrast, HybridRAG-BN combines retrieval with a fine-tuned verification stage [11]. The former helps distinguish context limitations from generation limitations; the latter emphasizes verification after retrieval. They address related problems through different designs.

For Civic.ai, the distinction between manually selected and automatically retrieved evidence motivated small diagnostic trials. Better generation with manually chosen passages does not establish that those passages will be selected automatically. We do not implement TraSe’s translation-and-selection architecture or train HybridRAG-BN’s verifier. Our selector instead uses service and action cues before generation, keeping the additional runtime mechanism relatively simple while leaving its decisions open to evaluation.

Long-context research also shows that information placement can affect model performance [12]. Our evidence-order trials examine a related practical concern using known civic questions. This motivates examining both context content and ordering; it does not prove that ordering alone explains every error or that our small diagnostic reproduces the published study.

Multilingual Models and Document Infrastructure. BGE-M3 supports dense, sparse and multi-vector retrieval [2]. We use only its dense mode; CivicRAG’s sparse component is BM25. This separates multilingual semantic representation from an interpretable lexical channel. ChromaDB provides persistent vector and document storage rather than a new ranking algorithm. These components support the dataset contribution but are not architectural novelty by themselves.

The Qwen3 and Qwen2.5 reports describe two of the model families used in our local experiments [4, 13]. We also evaluated the installed Llama 3 model. Our experiments concern the specific Ollama models requested in each run; local llama3 is not NextRAG’s Llama 3.1 variant. We selected Qwen3:8b for the principal paired comparison after project-level diagnostic runs, without claiming it is the best model for Bangla generally. Fixing the generator allows a clearer comparison of the retrieval-and-selection configurations.

Evaluation of Retrieval and Answers. RAGAS distinguishes context-supported statements from answer relevance [8]. An answer can repeat evidence accurately while discussing the wrong task. We therefore use saved contexts and responses for answer-level measures, and ranked chunk IDs with relevance labels for retrieval-level measures. Our custom binary context-relevance measure is identified separately from standard RAGAS metrics.

The current retrieval study includes assistant-reviewed pooled labels, not independently human-verified ground truth. Reference-answer similarity likewise requires reference answers; agreement between two generated outputs cannot substitute for them. These distinctions enable measurement without treating every available score as a correctness estimate.

Taken together, existing research provides established retrieval, generation and evaluation methods. The remaining project-specific gap is a structured, traceable collection of civic-service information and an implementation that preserves service conditions from documents to responses. Civic.ai addresses this gap through dataset development and an evaluated integration of existing methods, not a new general-purpose RAG architecture.

2.3 Summary of Key Findings

Three findings guide the design. First, multilingual semantic search and lexical matching address different retrieval needs, but their combination requires evaluation. Second, document structure and applicability matter alongside ranking: even a capable generator can receive the wrong procedure. Third, answer support, relevance and retrieval coverage describe different qualities.

These findings lead to the requirements in Chapter 3: preserve source structure, expose evidence, keep the pipeline modular and compare configurations under controlled conditions. They place dataset development at the centre of our contribution. The following chapters explain how these requirements are implemented and how the recorded experiments are interpreted.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

The needs-analysis survey informed the informational scope. Required-document guidance was selected by 62/77 valid respondents (80.5%), and payment/appointment information by 58/77 (75.3%). These preferences map to searchable instructions and source-grounded explanations. Demand for online-form help (64/77, 83.1%) and uploaded-document checking (56/77, 72.7%) was also recorded, but automatic submission and document authentication are outside the implemented requirements.

Functional Requirements. The system must accept a question with a selected domain, retrieve relevant passages, generate a Bangla response and expose the evidence supplied to the model. It must also support model selection and record answers, routes and source identifiers for comparison. A targeted clarification path handles recognized cross-domain aggregate requests; it is not a complete classifier for every out-of-domain question.

Non-functional Requirements. Traceability, maintainability and reproducibility guide the implementation. Original files must remain separate from derivatives, evidence must remain linked to stable IDs, and changes must be reversible. Language and incomplete-output checks support usability, while saved configurations permit reproducible analysis. Local response time is measured rather than treated as a guaranteed real-time service level.

Software Requirements. The implementation uses Python, domain data files, configuration files, BGE-M3 embeddings, ChromaDB, BM25, a BGE reranker and Ollama. The browser interface provides domain and model selection. Model weights must be available locally for offline execution; network access is needed for initial downloads and following

external source links. Qwen3:8b is explicitly requested in the main comparison, irrespective of default model names retained in other configurations.

Resource Requirements. Execution requires storage for the source archive, prepared corpus, indexes, model weights and experiment outputs. Available memory must support the selected embedding, reranking and generation components. Sequential batch execution and model sharing reduce concurrent resource demand. No minimum GPU specification is asserted because the project did not perform a hardware-capacity benchmark.

Table 3.1 connects the requirements with their implemented checks. This mapping sep-

Table 3.1: Requirement-to-implementation mapping for the final system.

Requirement	Mechanism	Verification boundary
Domain-scoped guidance	Separate configurations and collections	Not universal out-of-domain rejection
Evidence inspection	IDs, source metadata and answer contexts	Not claim-level citation validation
Bangla response	Prompt and language check	Does not guarantee fluent or correct wording
Local execution	Cached models and Ollama	Latency depends on resources
Reproducible comparison	Fixed inputs and recorded hashes	Questions reused during development
Reversible changes	Git revisions and saved artifacts	Only recorded files belong to a snapshot

arates an implemented mechanism from the broader quality it supports. For example, displaying sources satisfies traceability but cannot alone establish correctness.

3.2 Societal Impact

Civic.ai is intended to help citizens understand published procedures, not replace government services. Bangla explanations and inspectable evidence can make a dispersed collection easier to navigate. The survey indicates perceived demand for such assistance, but the completed experiments do not measure changes in office visits, financial burden or dependence on intermediaries.

The trust responses reinforce this distinction. Under a hypothetical verified-information condition, 42/76 respondents (55.3%) still preferred to recheck answers. A combined official-logo/website-link option was selected by 46/77 (59.7%). Its wording does not identify separate preferences for logos and links or justify official branding. We consequently emphasize source inspection without implying institutional endorsement.

Incorrect guidance can waste time or cause an applicant to prepare the wrong documents. Fees and procedures should therefore remain associated with their source conditions. Missing URLs and unresolved original-to-derivative matches are visible limitations of the dataset, not details to hide from users.

3.3 Environmental Impact

Local inference consumes electricity and hardware resources. Reusing pretrained models avoids conducting foundation-model training within this project, but does not establish a low-carbon system. Long inference and evaluation runs can still accumulate resource use.

We reuse saved answers for evaluation and preserve checkpoints instead of repeatedly regenerating completed batches. These practices avoid redundant work; their energy benefit was not measured. No power logs or carbon accounting were collected, so latency is reported as an operational measure rather than converted into unsupported environmental claims.

3.4 Ethical Issues

Source traceability is important because civic guidance can affect practical decisions. Original documents and derivatives are preserved separately, while identifiers and available URLs allow a response to be inspected. A correct source link does not certify that the derived text is accurate, current or applicable to every citizen.

The evidence-only prompt aims to reduce unsupported generation, and the pipeline can return clarification when no applicable evidence remains. Nevertheless, models can combine unrelated requirements or omit conditions. The system exposes incomplete-output warnings and selected passages rather than presenting these safeguards as a full factual verifier. Document dates and audit flags also identify where freshness and applicability require further checking.

Privacy is addressed differently for the corpus and survey. Public-service instructions form the retrieval corpus; respondent experiences are not added to it. Survey reporting uses aggregate counts from consenting adult submissions and excludes raw demographics, timestamps and free-text accounts from the distributed thesis artifacts. Four consenting under-18 submissions and two missing-age submissions are held out of the primary analysis. Participant identity and repeat participation cannot be reliably established, so counts refer to submissions rather than verified unique citizens.

Local inference avoids sending each question and passage to a hosted generator in the

recorded runs, but saved local logs can still contain sensitive user input. Their use in an operational deployment would require access and retention controls. No ethics approval or privacy certification is inferred from a consent checkbox or local execution alone.

3.5 Standards - if applicable

Encoding. UTF-8 storage and Unicode NFC normalization provide a consistent representation of Bangla and mixed-language text. Normalization standardizes encoding; it does not repair every OCR or spelling error.

Documentation. JSON/JSONL records, stable identifiers, source-relative paths and SHA-256 hashes support consistent data exchange and change detection. Missing URLs remain missing rather than being replaced with guessed addresses.

Software. Versioned code, configuration files and saved experiment artifacts support reproducibility. Matching the active corpus to its index is essential because retrieved IDs are resolved against the configured chunk map.

Government documents. Official titles, dates, forms and conditions are retained where present. Their preservation supports provenance but is not certification that a document is current or that redistribution is authorized. The study claims no formal government, security, accessibility or quality-management compliance audit.

3.6 Project Management Plan

We organized the work in successive stages: source preparation, a registration-focused implementation, extension to two further domains, generation diagnostics and controlled comparison. The sequence allowed changes to be tested without overwriting earlier results. Table 3.2 records milestones visible in the repository; dates identify recorded revisions, not the start or total duration of each activity.

The milestone record covers the final development period; earlier collection and initial implementation work precede these entries. Keeping the sequence tied to artifacts avoids reconstructing unrecorded working hours or assigning responsibilities retrospectively.

Version control was also a risk-control mechanism. Unsuccessful candidate changes were retained with their outputs, while rollback revisions preserved earlier behaviour. For comparison, the corpus, questions and generation settings were frozen so that later reporting did not silently change the experiment. Evaluation of saved answers was separated from model generation to reduce repeated local runs.

Table 3.2: Completed development milestones documented by repository revisions.

Recorded date	Milestone	Revision
19 Sep 2026	Passport/BRTA integration and three-domain audit preparation	ceba176
20–22 Sep 2026	Bangla policy, model audits and source-preserving cleanup candidates	61dd907, 143a552
23 Sep 2026	Context diagnostics, applicability selection and family-order correction	5389f7b, 27a1de4
23 Sep 2026	Saved local-model batches on the selected configuration	32228dc, 7ac52a1
24 Sep 2026	Matched Simple RAG run and needs-survey analysis	76a4537, c09fc42
25 Sep 2026	Retrieval-only measurements and completed saved-answer RAGAS evaluation	2ba8820, d32c107

3.7 Risk Management

Risks arise at each boundary between source material, retrieval and generation. Table 3.3 summarizes the controls already implemented and the issues that remain.

Table 3.3: Implementation-specific risks, controls and residual limitations.

Risk	Existing control	Residual issue
OCR corruption	Originals, source links and review flags	Long legal text not exhaustively checked
Wrong procedure	Domain scope, reranking and applicability cues	Similar headings or mixed passages remain
Outdated instructions	Date fields and audit flags	Legal currency not independently certified
Context omission	Whole-chunk budgeting and family ordering	Budget can omit relevant material
Incorrect generation	Source-grounded prompt and language check	No complete semantic verifier
Output truncation	Finish-reason warning	Warning does not complete the answer
Resource interruption	Checkpointed batch artifacts	Slow local inference remains
Evaluation bias	Matched inputs and disclosed label provenance	Reused questions and self-judging
Version mismatch	Corpus/configuration hashes and Git records	Rebuilds must match the intended snapshot

These controls reduce specific failure opportunities but do not remove the need to inspect errors. Keeping retrieval traces and answer contexts separate helps locate whether a failure originated before or after evidence selection.

3.8 Economic Analysis

The recorded generation runs use local Ollama rather than paid per-request inference. This removes hosted generation API charges for those runs, not all project costs. Hardware access, electricity, storage, document preparation and review remain resource inputs. The project did not collect a complete invoice, power or labour record, so an itemized monetary total cannot be reported reliably.

The economic trade-off is therefore described in terms of resources and workflow. Reusing pretrained models avoids a new training programme. Saved-answer evaluation avoids repeating generation, while checkpointed batches preserve completed work after interruptions. On the other hand, sequential local inference limits how many experiments can be completed in the available time.

For a measured run, an energy charge could be calculated as

$$C_{\text{energy}} = \frac{P_{\text{avg}} t}{1000} p, \quad (3.1)$$

where P_{avg} is average power in watts, t is elapsed hours and p is the tariff per kWh. This equation defines a cost model; it is not a measured cost result. Total attributable cost would also include hardware, storage and documented labour.

The survey provides a separate view of stated willingness to pay. Of 78 valid adult responses, 33 (42.3%) selected free-only use and 39 (50.0%) would pay if the service were fast and secure; six supplied other responses. These preferences are neither purchases nor a revenue forecast. Consequently, our economic contribution is a locally executable and inspectable workflow, not a demonstrated financial saving for citizens.

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

Our design follows the journey from a collected document to an inspectable answer. We first prepare a reusable knowledge base offline, then search it and generate a response when a question arrives. This separation allows source preparation, retrieval and generation to be examined independently, rather than rebuilding the entire system for every question.

The architecture figures in Section 4.2 show how source content remains connected to generation through chunk identifiers. Index construction does not train an LLM. Dense search, BM25, fusion, reranking, applicability selection and generation are shared across the three domains. Additional registration-specific adjustments are distinguished from shared reranking. Legacy controlled-answer branches remain in the repository but do not generate answers in the evaluated selected-evidence route.

To connect these stages, we use the saved question BRTA-এর ড্রাইভিং পরীক্ষায় কী কী থাকে? (What does the BRTA driving test include?) as a running example. The record is identified as `brta_05_test`. It illustrates an actual execution, not a reference answer or a demonstration that every selected passage is relevant.

4.2 Preliminary Design or Design (Model) Specification

Figure 4.1 presents the complete implemented architecture using a corrected export of the supplied draw.io design. All three stages and their connecting arrows remain together in one figure. Figure 4.2 separates the two evaluation designs from the online chatbot.

In Figure 4.1, the first panel starts from source files and prepared derivatives. Each chunk supplies search text for BGE-M3 and BM25, while evidence content stays linked by ID.

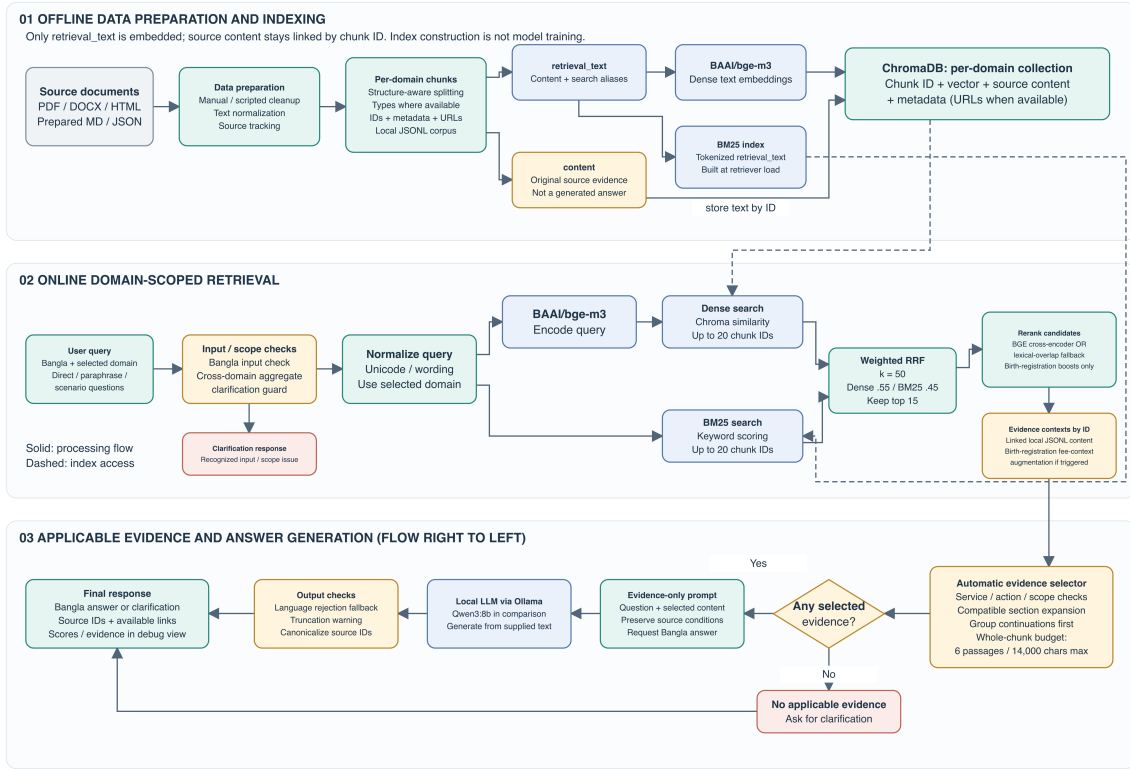


Figure 4.1: Complete CivicRAG architecture: offline preparation, domain-scoped retrieval, and evidence-based generation. All inter-stage connectors are retained. Solid arrows show processing; dashed arrows indicate index access. The final panel flows right to left.

Only search text is embedded; Chroma stores the vector, linked document and metadata [2]. BM25 is constructed from tokenized text when the retriever loads, rather than stored as a Chroma vector.

The second panel follows a question through domain-scoped retrieval. After input checks and normalization, semantic and lexical searches produce at most 20 results each. Weighted RRF retains up to 15 fused candidates, and the shared reranker reorders them [3]. The local chunk map supplies evidence throughout retrieval; the evidence-context box denotes preparation for selection, not the first moment the code accesses text. Registration-only score adjustments and fee supplementation are conditional additions, not separate retrieval methods.

The bottom panel of Figure 4.1 runs from right to left. A rule-based selector checks service/action scope and orders compatible sections before applying the whole-chunk budget. With no selected evidence, the system requests clarification. Otherwise, the question and source text enter the generator; vectors and search aliases are not factual answer content. Source-ID canonicalization occurs inside the generator wrapper, while language and truncation checks occur in the pipeline. The grouped output-check box summarizes these operations rather than defining a strict internal order. There is no second model verifying

every final claim.

Figure 4.2 has two independent branches. The first compares saved answers from matched Simple RAG and CivicRAG configurations. The same prompt means the same generation instructions, not the same passages or selector. The second compares BM25, dense search and RRF before reranking or selection. RAGAS scoring and passage-label review occur outside the citizen-facing request path. The diagram reports their evidential limits; the retrieval-only study does not measure complete chatbot accuracy.

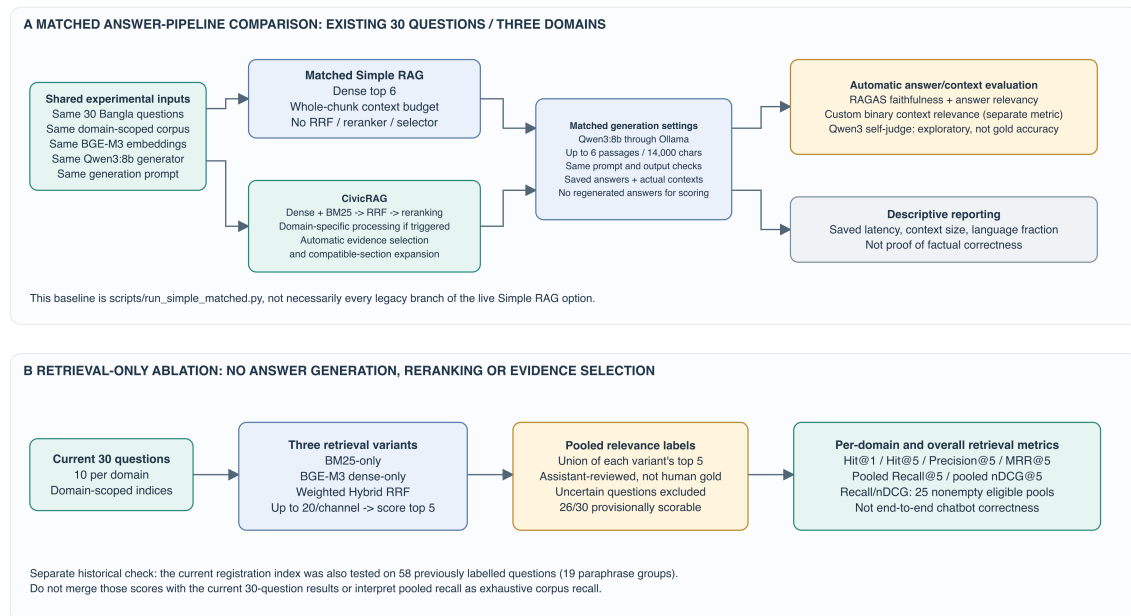


Figure 4.2: Evaluation boundaries. Panel A compares matched answer pipelines; Panel B evaluates retrieval before reranking and evidence selection. Assistant-reviewed labels and local self-judge scores are distinct from independent ground truth.

The initial design emphasized birth registration, hybrid retrieval and controlled answer paths. Later work extended the corpus to passport and BRTA, added applicability selection and changed the main generator comparison to Qwen3. The final reported route therefore differs from the old diagrams that send every answer through a generic safe-path box.

The system is not an autonomous agent that performs government actions. Its task is single-turn retrieval and evidence-conditioned text generation. A domain is selected before retrieval. A targeted cross-domain aggregate guard handles certain multi-service questions, but comprehensive intent recognition is not claimed. Neither the lexical rules nor the selector is a separately trained classifier.

We considered lexical-only retrieval, dense-only retrieval and their hybrid combination. BM25 provides exact-term matching with little model computation, while dense retrieval accommodates semantic similarity. Hybrid retrieval makes both signals available, at the cost of additional ranking and potentially distracting candidates. The end-to-end alternatives are the matched dense-only Simple RAG baseline and CivicRAG's hybrid, reranking

and selected-evidence configuration.

We retained the hybrid configuration as the design under investigation because exact service terms and semantically related wording both occur in the corpus. This is an engineering rationale, not a claim that hybrid retrieval won every comparison. The earlier registration-focused retrieval results favoured dense-only search on several reported measures. Later relevant-only and evidence-order diagnostics showed why context applicability also matters, motivating selection after retrieval. Those diagnostics do not establish the selector’s benefit on arbitrary questions. The bundled comparison and retrieval-only tests evaluate these choices; a selector-off answer ablation was not completed.

Table 4.1 explains the principal decisions and their trade-offs. The table separates why

Table 4.1: Design decisions and implementation-specific justifications.

Choice	Reason for selection	Trade-off or boundary
Structured chunks	Preserve conditions and source links	Depends on the quality of derivatives
BGE-M3	Multilingual semantic representation	Embedding similarity is not applicability
ChromaDB	Persistent local vectors and linked records	Index and chunk map must remain consistent
BM25	Exact service terms and form labels	Sensitive to wording and token overlap
Weighted RRF	Combine ranks on different score scales	Fixed weights are not proven optimal
BGE reranker	Joint query-passage scoring	Extra inference cost; fallback differs
Applicability selector	Screen scope and preserve related sections	Heuristic decisions can omit useful text
Ollama / Qwen3:8b	Local serving and a fixed comparison generator	Slow inference and imperfect Bangla remain

a component was chosen from whether it improves a measured outcome. Keeping these questions distinct supports a fair comparison in Chapter 5.

4.3 Data Collection -(If Applicable)

Needs-Analysis Questionnaire. The team supplied three CSV questionnaire exports, each containing 46 columns. They cover demographics, experiences with birth registration, passport and BRTA services, and a common nine-item block concerning a prospective AI assistant. The three versions place the service-specific blocks in different orders. Analysis aligns those blocks before checking identical submissions and uses column positions because several free-response headers repeat. The survey is a requirements input,

not additional retrieval evidence, and is not ingested into the chatbot’s knowledge base.

Only affirmative-consent records with a declared adult age group enter the main descriptive analysis. This conservative inclusion rule was applied when preparing the thesis analysis; it is not claimed to have been preregistered. Missing responses are excluded separately for each item, not interpreted as rejection. Percentages are computed as $100c/n_v$, where c is the number selecting a response and n_v the number answering that item. Multiple-selection items count each explicitly selected option once per submission, so their percentages need not sum to 100. A free-text “all options” response is retained in the denominator but not expanded into checked options. Open-text answers are not quoted publicly or automatically coded as validated themes.

The exported consent field establishes an affirmative response, not independent ethics approval. Recruitment method, invitation count, repeat-participation controls and consent arrangements for minors were not supplied. Accordingly, a response rate and nationally representative estimates cannot be calculated. The BRTA website question also combines a Bangladesh address with `rsta.gov.bt`; that item is not used to claim verified BRTA website usage. The main findings use the identically positioned common AI block and are reported in Chapter 5.

Source Families and Original Preservation. The birth registration collection contains structured JSON and prepared Markdown derived from FAQs, laws, rules, fees, correction processes and application guidance. Passport and BRTA were supplied as original-file directories and separately prepared Markdown. Passport derivatives are registered from `cleaned_md`; BRTA derivatives from `brta` and `fahim_brta`. Directories explicitly marked original are reference materials, not duplicate text automatically added to the answer corpus.

The register preserves document IDs, relative paths, snapshots, hashes, candidate original-file matches, inclusion history and chunk IDs. Candidate original matches do not certify textual fidelity. Some matches remain unresolved and are flagged. The distinction between collected originals and derivatives avoids counting the same information twice as independent evidence.

The corpus is a retrieval resource, not a newly annotated gold-answer benchmark. Its novelty is the organization of selected civic-service material into a source-traceable, machine-usable representation spanning the three domains. It is neither a census of all government documents nor a guarantee of current legal validity.

The collected source files and team-prepared derivatives are separate layers. Document-understanding tools such as Docling can convert structured PDF content into Markdown [5], but the supplied records do not identify that tool as the converter for our files. The

reproducible preparation described here begins with the archived derivatives and their available original-file associations.

4.3.1 Data Cleaning

Figure 4.3 summarizes how source preservation, normalization and structural preparation lead to the active retrieval corpus. The figure separates immutable reference materials

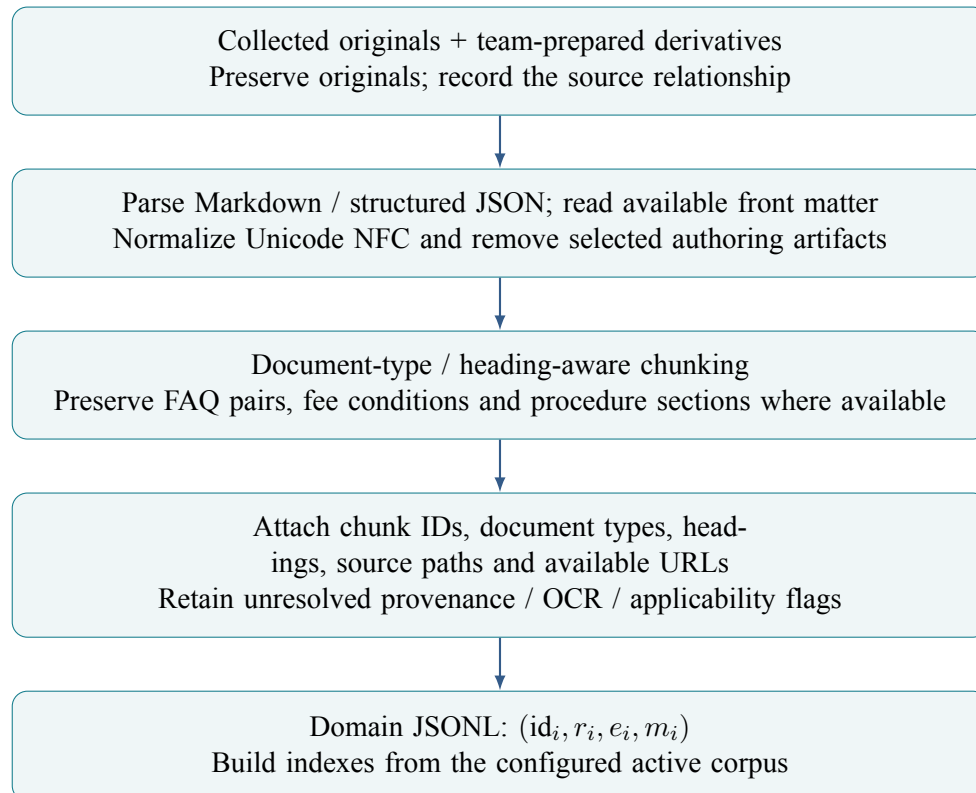


Figure 4.3: Source-preserving preparation of the active corpus: original materials remain available while normalized derivatives are organized into identifiable retrieval units. Separate cleanup candidates are not substituted for the evaluated corpus.

from transformations applied to derivatives. The conversion stage and subsequent cleaning are not interchangeable: a structured converter such as Docling can recover layout [5], while our confirmed ingestion normalizes and partitions the already prepared text.

The active passport/BRTA ingestion parses simple front matter, normalizes NFC text, removes explicit authoring citation markers and selected invisible characters, and tracks headings. It preserves malformed or unrecognized material instead of making broad deletions. The work does not substantiate a universal automatic OCR repair or Bijoy-to-Unicode conversion stage for every supplied file; such claims from illustrative diagrams are not carried forward.

Separate cleanup candidates remove recognized authoring preambles and boilerplate and

preserve larger structural units. Their manifests record changes and review flags. Those candidates must be distinguished from the evaluated corpus. At the audited snapshot, the active domain configurations still reference the heading-aware v2 passport/BRTA files. A later cleanup directory alone is not evidence that its contents were activated in ChromaDB or used in the saved answers.

4.3.2 Data Transformation

Document-type-aware transformation retains question-answer pairs, headings, fee rows and process sections where the source permits. A chunk should keep its governing condition with its answer. Splitting a heading from its body can make retrieval apparently relevant while the actual content answers a different question. Heading-aware ingestion was introduced to reduce this failure, but long unstructured legal OCR remains difficult.

Each chunk has the form

$$c_i = (\text{id}_i, r_i, e_i, m_i), \quad (4.1)$$

where e_i is evidence content, r_i is search text and m_i is metadata. Search text may include aliases or section descriptors. These additions help search; they are not source facts and are not meant to become answer evidence. No query-specific reference answer is inserted into the generator merely because it resembles a test question.

Metadata retains Bangla or English titles as supplied, together with consistent field names for the domain, document identifier, section, source and audit flags. This is bilingual metadata preservation, not automatic translation of every record. For the running BRTA example, a passage about a driving test remains associated with its own section and document even when other licence-related passages share the same query terms.

4.3.3 Data Integration

Figure 4.4 illustrates how the search vector, original evidence and metadata remain joined by one identifier. Each domain has its own configured JSONL corpus and Chroma collection. Metadata preserves service context and available source URLs. A stable chunk ID joins vector results back to the source record. The runtime dense retriever obtains IDs from Chroma and reconstructs retrieval results from the in-memory chunk map, making the active JSONL and index consistency essential.

In `scripts/build_index.py`, embeddings are computed from `retrieval_text`, whereas Chroma’s documents contain content. The versioned ingestion script follows the same storage separation. Therefore answer evidence is stored alongside a vector; it is not itself an answer produced by Chroma. Chroma is a retrieval store, and the LLM produces the

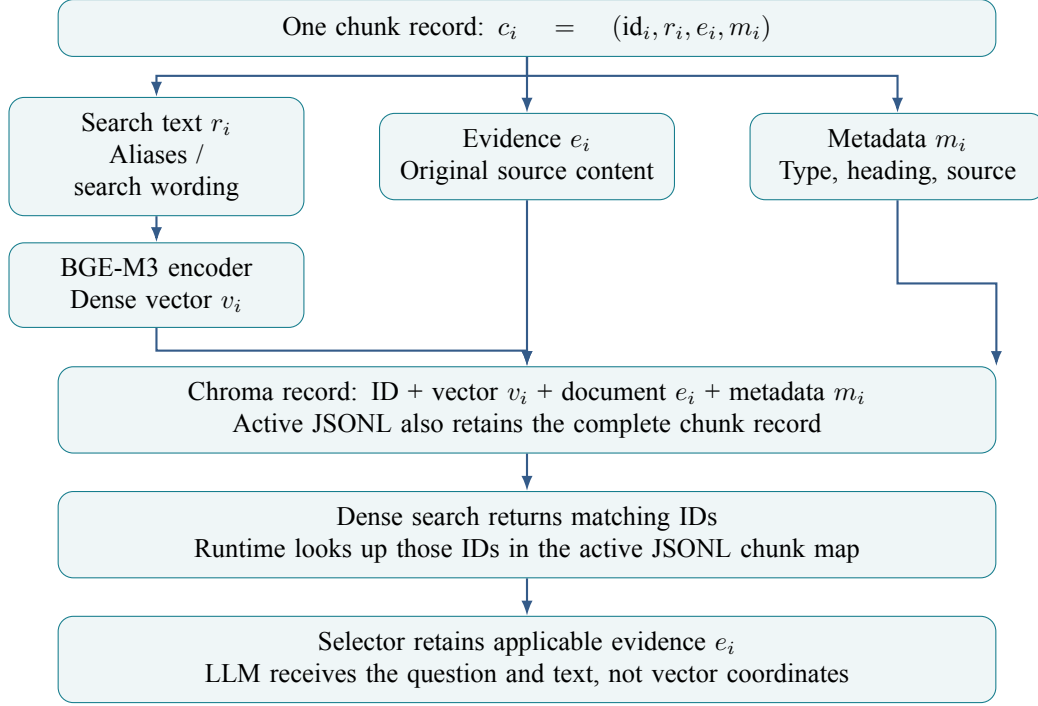


Figure 4.4: How answer evidence remains connected to a vector without being generated by the database. BM25 is a separate index of r_i , omitted here to isolate the dense-storage link. All arrows entering storage denote stored fields, not additional embeddings.

response only after evidence selection.

Figure 4.4 makes this relationship explicit. If I_q is a list of retrieved chunk IDs and M is the active chunk map, then the corresponding evidence is obtained by

$$E_q = [M[i].content : i \in I_q]. \quad (4.2)$$

This is an identifier lookup, not a comparison between a database-generated answer and a reference answer. The subsequent selector can remove candidates or append compatible family members; therefore the final supplied evidence need not equal the first retrieved list.

4.3.4 Data Reduction

Reduction is treated conservatively. FAQ duplicate handling and chunk boundaries can reduce repeated text, but semantic deduplication of the complete three-domain corpus has not been independently established. Original archives are retained. Passport and BRTA registers contain 30 and 132 derivative documents respectively; the active snapshot includes chunks from all of those registered derivatives. This statement does not imply that every original file has a resolved one-to-one derivative match.

Chunking reduces the size of each retrieval unit rather than necessarily reducing the total amount of stored text. Known authoring artifacts are removed, while headings and conditions are retained where possible. This helps retrieval target a procedure instead of presenting an entire legal document to the model. Over-aggressive reduction would remove exceptions or qualifications, so flagged material is not silently discarded merely because it is inconvenient to process.

4.3.5 Summary of Preprocessed Data

The corpus table reports the active, configured source units and chunk counts. Figure 4.5 visualizes their distribution across domains. Figure 4.5 shows the corpus imbalance:

Table 4.2: Active corpus inventory computed from configured files. Source units are prepared documents, not independent facts.

Domain	Source units	Registered originals	Chunks
Birth registration	12	Not comparable	187
Passport	30	27	361
BRTA	132	133	3161

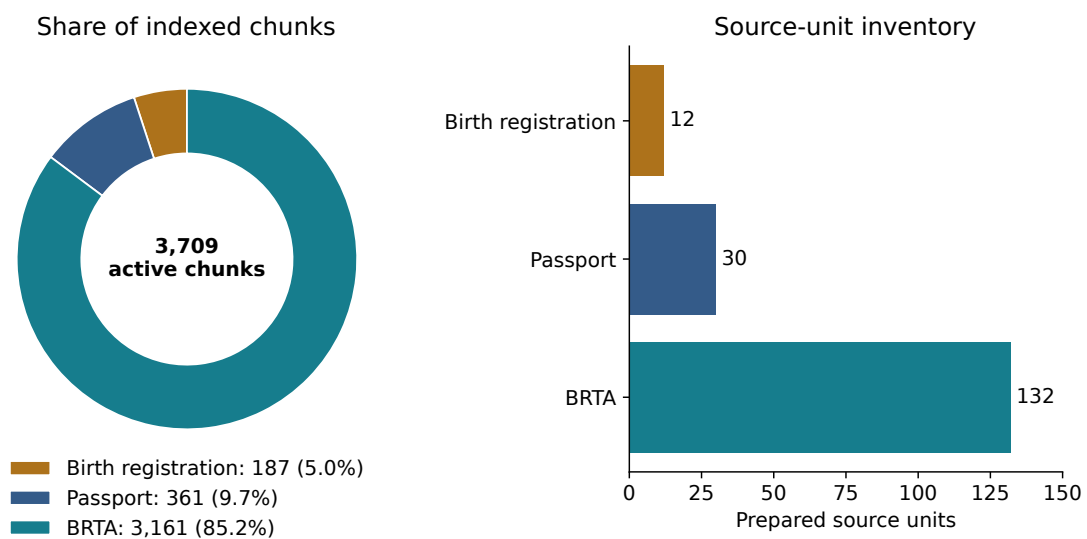


Figure 4.5: Active corpus composition from the frozen evidence manifest: 3,709 chunks and 174 prepared source units. Chunk share reflects document length and splitting, not demand, accuracy or unique facts.

BRTA accounts for 85.2% of chunks, passport for 9.7% and birth registration for 5.0% (rounded independently). Domain-scoped retrieval prevents this corpus-size imbalance from directly mixing unrelated services in one query. It does not establish equal retrieval difficulty or equal source coverage across domains. The table is computed from the configured JSONL files, not from unrelated candidate directories. For the older birth registration

corpus, source units are counted using its source-path metadata. That count is not equated to a unique original-PDF count. Across domains, chunk counts are an indexing property rather than the number of independent facts or questions.

Table 4.3: Selected unresolved active-corpus flags, counted once per derivative document. These counts may overlap.

Domain	Flag	Documents
Passport	no web address	26
Passport	original match unresolved	4
Passport	dated document review applicability	1
Passport	long document check ocr	0
BRTA	no web address	109
BRTA	original match unresolved	6
BRTA	dated document review applicability	21
BRTA	long document check ocr	8

Review flags count documents in the passport/BRTA registers or their associated chunk metadata, as indicated in the table. Multiple flags may apply to one document. A missing URL does not prove that a source is false, while a present URL does not prove that the derived text is correct. The flags identify where provenance and applicability need further review.

4.4 Implementation of Selected Design

The implementation follows Section 4.2 through explicit modules rather than an autonomous-agent framework. The index builder constructs Chroma records; HybridRetriever implements retrieval and fusion; HybridReranker performs reranking; and select_evidence implements applicability and budgeting. CivicRAGPipeline coordinates these stages and response routes. Finally, OllamaAnswerGenerator calls LangChain’s ChatOllama integration, while separate evaluation scripts consume saved outputs. LangChain is an adapter here, not another retriever or a trained model.

The following stages describe the implementation used by the saved comparison. For the running example, the user selects BRTA and submits a Bangla question. The interface applies its Bangla-oriented input policy; the pipeline normalizes the query and uses only that domain’s configured collection. This does not require the question to match a stored FAQ exactly. The recognized cross-domain aggregate guard can request a narrower question before retrieval, but it is not a general intent classifier.

Dense Embedding and Search. With encoder f_θ , normalized vectors are

$$v_i = \frac{f_\theta(r_i)}{\|f_\theta(r_i)\|_2}, \quad v_q = \frac{f_\theta(q)}{\|f_\theta(q)\|_2}. \quad (4.3)$$

For unit vectors, squared Euclidean distance satisfies

$$\|v_q - v_i\|_2^2 = 2 - 2v_q^\top v_i. \quad (4.4)$$

Thus distance and cosine induce the same exact ranking for normalized vectors. The index construction does not explicitly override Chroma’s distance configuration; the thesis therefore does not assert that a cosine collection was manually configured. Approximate index search can differ from an exhaustive mathematical ranking.

BGE-M3 supplies the embedding model [2]; Chroma stores the vectors and linked documents. The implementation uses local cached model files and CPU embedding configuration. No corpus-specific gradient training occurs during this step.

For the driving-test question, dense search compares the normalized query representation with the BRTA passage vectors. It seeks semantic proximity, not an answer string stored in the database. The linked evidence text is recovered after matching chunk IDs.

BM25 Sparse Search. BM25 indexes tokens from r_i . Tokenization normalizes NFC, removes zero-width characters and extracts lower-case word/Bangla token sequences. The standard scoring structure is

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \frac{f(t, d)(k_1 + 1)}{f(t, d) + k_1(1 - b + b|d|/\bar{L})}. \quad (4.5)$$

Here $f(t, d)$ is term frequency, $|d|$ is token length and $\bar{L}$ is mean document length. The code instantiates `BM25Okapi` without explicit parameter overrides, so the installed library defaults determine k_1 , b and its IDF-floor behaviour [6]. Lexical matching is a separate channel; it does not construct Chroma’s vectors.

For the same question, BM25 emphasizes overlapping words such as driving and examination in the prepared text. It can retrieve a passage missed by semantic search, but may also retrieve another kind of licence examination. The subsequent stages must address that ambiguity.

Weighted Reciprocal Rank Fusion. The implementation retrieves at most 20 candidates from each channel, omitting BM25 results with non-positive scores. It assigns one-

based ranks and computes

$$S(d) = \frac{0.55 \mathbf{1}[d \in D]}{50 + r_D(d)} + \frac{0.45 \mathbf{1}[d \in B]}{50 + r_B(d)}. \quad (4.6)$$

A missing-channel term contributes zero. As an illustrative calculation, a chunk at dense rank 2 and BM25 rank 5 receives $0.55/52 + 0.45/55 \approx 0.01876$. This is a rank-combination value, not confidence in factual accuracy. The fused list is limited to 15 before reranking. The weights are implementation settings; this thesis does not claim they are optimal for the dataset.

The example calculation is explanatory rather than a reconstruction of the saved question’s scores. In the actual trace, channel markers record whether a candidate came from dense search, BM25 or both. Fusion makes a combined candidate list available; it does not yet determine the final prompt.

Reranking and Domain-Specific Logic. The configured cross-encoder scores query/search-text pairs. If loading fails, the code can add a lexical-coverage adjustment of $0.05 |T_q \cap T_d| / \max(|T_q|, 1)$ to the incoming score. The saved primary CivicRAG candidate traces contain cross-encoder markers, establishing that the cross-encoder route was used in those records rather than merely being configured.

Birth registration-specific boosts and evidence augmentation remain part of the selected route. These are not universal learned mechanisms for passport and BRTA. This asymmetry should be disclosed when interpreting domain differences. An isolated ablation would be required to attribute an observed change to those rules.

Shared Reranking versus Registration-Specific Adjustments. The distinction is explicit in `src/pipeline.py`: `birth_death_rules` is true only for the `birth_death_registration` domain, and that flag is passed to the reranker’s `domain_boosts` option. The cross-encoder itself is shared; passport and BRTA do not skip reranking. Its base score may be written as $s_{\text{CE}}(q, r_i) = g_\phi(q, r_i)$, where the pretrained cross-encoder jointly processes the query and searchable passage. For a domain d , the implemented adjustment has the form

$$s_d(q, c_i) = s_{\text{base}}(q, c_i) + \mathbf{1}[d = \text{birthregistration}] \Delta_{\text{registration}}(q, c_i). \quad (4.7)$$

Here s_{base} is the cross-encoder score, or the documented fallback score when that model is unavailable. $\Delta_{\text{registration}}$ is the sum of applicable hand-written adjustments, not a learned probability. For example, an application-process record receives an increment of 0.08 for a detected registration how-to query; additional heading conditions can further alter that

score. These increments do not have a calibrated confidence interpretation.

Separately, `_augment_contexts` can append an existing correction-fee row for a detected date-of-birth or other data-correction query, provided its ID is not already present. It neither writes a new fact nor creates a synthetic answer. This registration-specific supplementation is different from the shared selector’s compatible-family expansion, which can also operate on passport or BRTA evidence.

These rules reflect the earlier registration-focused implementation and its document-type schema. Applying the same conditions unchanged to other domains would be inappropriate: a birth-registration correction fee is not a passport or driving-licence correction fee. Equivalent domain rules were not implemented for passport and BRTA. This is an engineering asymmetry to disclose, not evidence that the extra rules are necessary or beneficial in every case. No pipeline change is made by documenting it.

Accordingly, the BRTA running example receives shared reranking but skips the additional registration rules. The saved trace includes cross-encoder markers, while the highest-ranked initial candidate concerns a professional-licence dope test. This illustrates why a high relevance score is not sufficient to determine whether a passage directly answers the intended examination question.

Applicability Selection and Context Budget. The selector introduced during incremental development uses service, action, product and location vocabularies. It prefers section scope over broad document titles and can reject a passport-collection passage for an application question. It also screens heading-only fragments, handles explicitly flagged historical alternatives and expands compatible fee/checklist families.

The selector is implemented in `src/generation/evidence_selection.py`; it is neither a second LLM nor a human choosing passages during normal operation. It retains at most six whole chunks within 14,000 characters. A chunk that does not fit is skipped rather than silently cut. A later ordering fix groups compatible family members near the first eligible anchor before applying the budget. Saved replay on 30 candidate sets changed the selected IDs for the passport-fee question while leaving the other 29 selections unchanged. This establishes the scope of that replay, not a general accuracy improvement.

For eligible candidates, the selector’s initial priority is

$$h(q, c_i) = 3A_i + S_i + 2T_i, \quad (4.8)$$

where A_i indicates an overlapping action label, S_i an overlapping service label and T_i a fee/document-type match. Product, location and action conflicts can exclude a candidate before ordering. Ties retain the incoming candidate order. Historical-alternative filtering,

checklist handling and compatible-family grouping are additional procedural steps, so this equation alone is not the entire selection algorithm.

For the resulting ordered list, the greedy whole-chunk selection obeys

$$|E| \leq 6, \quad \sum_{c_i \in E} \text{len}(e_i) \leq 14,000. \quad (4.9)$$

The second bound uses Python string length in characters, not tokenizer tokens. This is a budget constraint, not a globally optimized evidence set. Skipping an oversized chunk can allow a later shorter chunk to enter the prompt.

An unknown or broad query can still retain unrelated material. Lexical applicability is not entailment, and a passage may contain mixed procedures even when its heading matches. The selector’s decisions and expanded IDs are therefore retained in a trace. Removing the selector remains a proposed ablation, not a completed experiment in this manuscript.

For `brta_05_test`, the saved selector trace identifies the service as licence, finds no explicit action label, expands no family members and retains six chunks containing 5,770 characters. The selection still includes neighbouring licence topics. This concrete trace shows both what the function does and why its output is not equivalent to manually verified relevant-only evidence.

Answer Generation and Safety Behaviour. Qwen3:8b receives the normalized question and selected evidence through Ollama [4]. The comparison uses temperature 0.05, top- p 0.9, output limit 1,200 tokens, repetition penalty 1.18, repeat window 256 and selected-route context window 8,192. Qwen3 thinking is disabled. The prompt requests evidence-grounded Bangla responses and preservation of service conditions.

The current comparison bypasses legacy controlled extraction and question-category templates. If no applicable evidence remains, it returns a clarification/insufficient-information response. A heuristic language check can replace unsuitable-language output. A length finish reason produces an explicit incomplete-output warning. These are operational safeguards, not a complete factual verifier.

The answer’s Sources footer identifies supplied context IDs, not verified supporting citations for each claim. The saved `raw_generation` field is produced after the generator’s existing response canonicalization and should not be described as untouched token-level model output. Candidate sources and actual `answer_contexts` are stored separately to avoid evaluating unused passages as if the model had seen them.

In the running example, the recorded route is `llm_selected_evidence` and generation ends with a normal stop reason. The response lists লিখিত পরীক্ষা, মৌখিক পরীক্ষা and ব্যবহারিক পরীক্ষা, followed by a website suggestion. The complete saved response, in-

cluding its original wording and all six supplied IDs, is retained in the experiment artifact. This excerpt illustrates the transition from evidence to an answer; it is not independently verified government guidance or a claim that all six passages support every statement.

Matched Simple RAG Baseline. The isolated runner `scripts/run_simple_matched.py` performs dense-only top-six retrieval. It does not call BM25 fusion, reranking, applicability selection, sibling expansion or controlled answers. It applies the same whole-chunk character budget, selected-route generation prompt, Qwen3 settings and language/truncation handling as the comparison configuration. The `selected_evidence=True` generator flag chooses the shared prompt; it does not invoke the selector inside the baseline.

Table 4.4: Controlled and varied elements of the main comparison.

Element	Simple RAG	CivicRAG
Corpus and questions	Same	Same
Generator and prompt	Qwen3, matched	Qwen3, matched
Dense retrieval	Top 6	Top 20
BM25 and RRF	Absent	Present
Reranker	Absent	Present
Applicability/expansion	Absent	Present
Final context limit	6 / 14,000 characters	6 / 14,000 characters
Controlled answer bypass	Absent	Absent

The matched settings isolate a bundled retrieval-and-selection difference while keeping corpus access and generation conditions comparable. They do not independently establish the contribution of RRF, reranking or selection. Retrieval-only experiments examine BM25, dense search and RRF before these later stages; their ranking metrics answer a different question from the end-to-end answer comparison.

Chapter 5

Result Analysis

5.1 Performance Evaluation

Needs-Analysis Results: Separate from System Performance. The three survey exports contain 66, 13 and 7 submissions, respectively (86 total). One explicit non-consenting submission is excluded. Of the 85 affirmative-consent submissions, four report an age below 18 and two omit age. The primary analysis therefore includes 79 adult submissions. No identical complete submission was found after aligning the service blocks, but this does not establish unique participant identities. The analysis is reproducible from the private archive using `scripts/analyze_needs_survey.py`; public artifacts contain aggregate counts, inclusion rules and an archive hash only.

The sample is concentrated among younger adults and men: 63/79 (79.7%) report age 18–25, and 73/79 (92.4%) report male gender. Recruitment and invitation counts are unavailable. Findings describe this responding sample and must not be extrapolated to all Bangladeshi citizens. They are not results from the 30-question chatbot experiment.

Table 5.1: Needs-analysis responses among consenting adults. Denominators exclude blank answers to each item.

Response	Count / valid responses (percentage)
Would use the proposed AI assistant	55/77 (71.4%)
Would not use it	13/77 (16.9%)
Unsure about using it	9/77 (11.7%)
Would recheck even verified information	42/76 (55.3%)
Bangla would be more comfortable	19/77 (24.7%)
Either Bangla or English acceptable	50/77 (64.9%)
Would recommend if it works well	67/76 (88.2%)

Among valid willingness responses, 55/77 (71.4%) would use the prospective assistant, 13/77 (16.9%) would not and 9/77 (11.7%) were unsure. The two missing responses are

not counted as either acceptance or rejection. For trust in verified information, 30/76 (39.5%) answered yes, 4/76 (5.3%) no and 42/76 (55.3%) would recheck. Positive willingness and cautious trust can coexist: the questionnaire supports the need for helpful information with verification opportunities, not unrestricted reliance on AI.

Figures 5.1 and 5.2 visualize these preferences and the requested kinds of assistance.

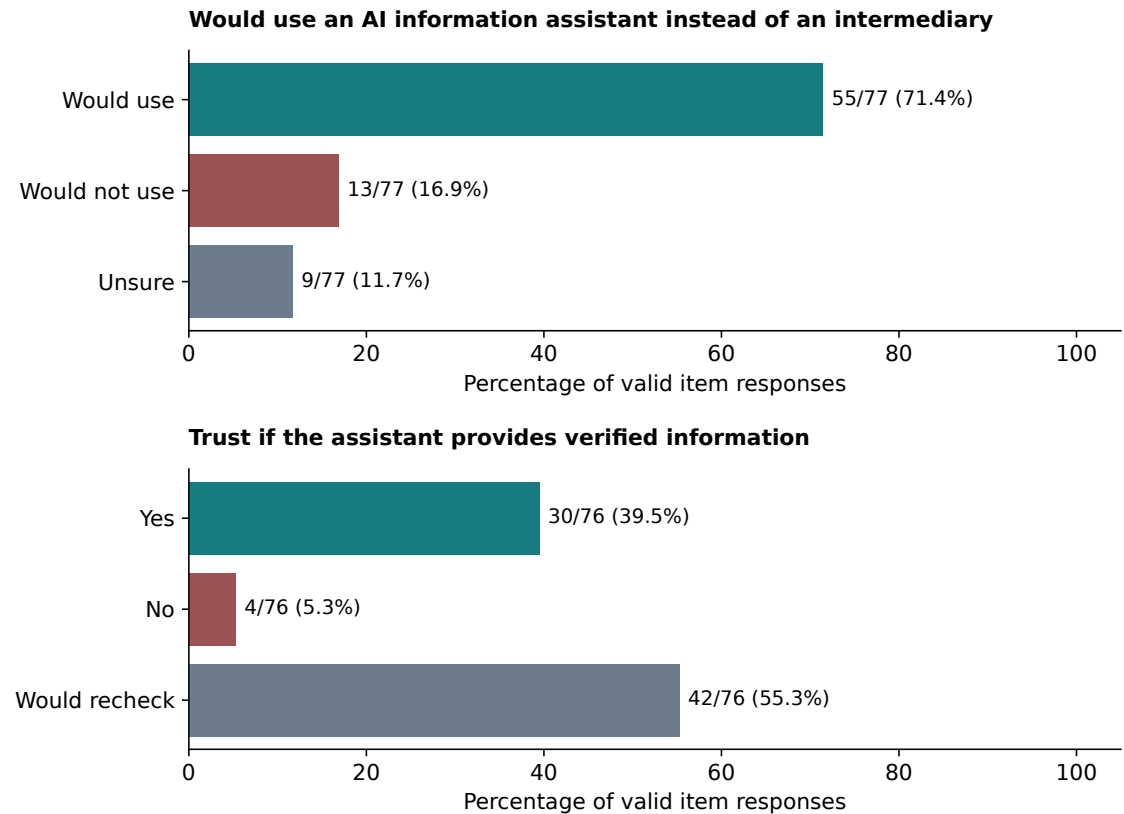


Figure 5.1: Stated willingness and conditional trust in the needs-analysis questionnaire. Only consenting adult submissions are included; denominators differ because of missing answers.

Required-document information (80.5%) and payment/appointment information (75.3%) align with the implemented question-answering scope. Form-completion assistance (83.1%) and uploaded-document checking (72.7%) indicate additional expectations beyond that scope. This distinction prevents reporting questionnaire demand as proof that the system already satisfies every requested function.

The language item does not establish a majority preference for Bangla-only operation. Nineteen of 77 respondents (24.7%) said Bangla would be more comfortable, 50 (64.9%) accepted either Bangla or English and eight (10.4%) answered no. Bangla-focused evaluation is therefore a project scope decision consistent with part of the expressed need, rather than a requirement endorsed exclusively by most respondents. Conditional recommendation was positive for 67/76 (88.2%), but the question assumes the assistant works well; it is not satisfaction measured after using CivicRAG.

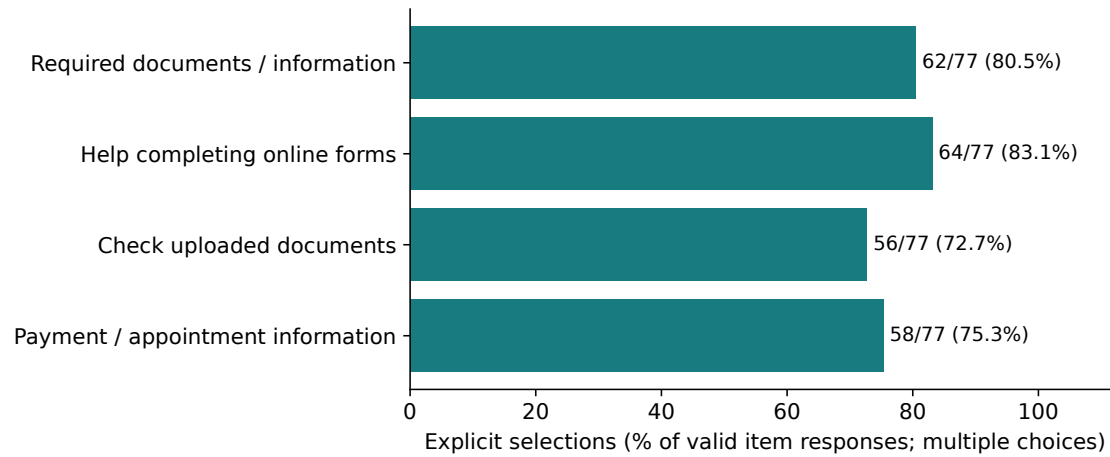


Figure 5.2: Requested assistance, based on explicit selections among 77 valid responses. Multiple choices were allowed. One “all options” response is not expanded, so explicit-choice counts are conservative. Requested features are not necessarily implemented capabilities.

Experimental Design. The primary comparison uses the same 30 Bangla questions and domain-scoped corpus for Simple RAG and CivicRAG. Each system produced one saved answer per question using Qwen3:8b, the matched generation prompt and the same context and output limits. Simple RAG uses dense retrieval without fusion, reranking or applicability selection; CivicRAG includes these stages as described in Section 4.2. The comparison measures the combined configuration difference, not the isolated contribution of each component.

There are ten questions each for birth registration, passport and driving-licence services. All are labelled direct in the saved plan. They concern requirements, fees, time, application, correction, renewal, replacement and status. They were reused during development, so these are development-set results rather than estimates on an unseen benchmark. The source answers and hashes are preserved in the evaluation archive. No answers were re-generated for scoring or thesis preparation.

Operational Outcomes. Table 5.2 reports saved latency and context size. Both systems completed 30 requests; completion is not a correctness label. CivicRAG supplied fewer passages and characters on average. Its saved mean latency was also lower, but machine load, thermal conditions and initialization were not controlled across the two runs. Therefore the table does not establish a hardware-normalized speed advantage. Bangla-letter fraction measures script use after removing the source footer; it is not a fluency or correctness score.

Table 5.2: Operational measurements for the matched 30-question runs. Times are seconds; runs occurred separately.

Measure	Simple RAG	CivicRAG
Completed requests	30	30
Mean elapsed time	193.699	181.823
Median elapsed time	181.060	161.845
95th percentile	319.910	285.100
Mean supplied passages	6.000	4.900
Mean context characters	5218.900	3948.033
Mean Bangla-letter fraction	0.935	0.933

Completed RAGAS and Custom Context Evaluation. The evaluator uses RAGAS 0.2.15 and actual supplied answer contexts, not every retrieved candidate [8]. There are $30 \times 2 \times 3 = 180$ completed metric scores, all valid, with no missing final scores. The appended Sources footer is removed before evaluation; response wording is otherwise preserved.

Faithfulness estimates support for the response statements in the supplied context:

$$F = \frac{\sum_{j=1}^m \mathbf{1}[a_j \text{ is supported by context}]}{m}. \quad (5.1)$$

Here a_j is a judge-extracted statement. This tests textual support, not current legal validity. Answer relevancy reconstructs three questions from a response and compares their BGE-M3 embeddings with the original question, with the metric’s noncommittal-response handling. It is not similarity to a reviewed reference answer.

The third measure is a custom binary RAGAS AspectCritic. It asks whether most supplied context concerns the requested service, procedure and conditions, instructing the judge to ignore the response. Each context set receives 0 or 1. Its mean is therefore a share of positively judged context sets, not passage-level context precision or recall.

The judge is Qwen3:8b, also the answer generator. The evaluation uses temperature zero, seed 20260924, non-thinking mode, a 16,384-token context window and a 4,096-token output allowance. Stock evaluation prompts are applied to Bangla content. This completed computation remains an exploratory self-judge evaluation: a fixed seed and valid output format do not make the judge independent or its scores equivalent to factual accuracy.

Table 5.3 presents the completed means, and Figure 5.3 visualizes the same values. Simple RAG has higher mean faithfulness (0.7983 versus 0.7323) and slightly higher answer relevancy (0.8460 versus 0.8375). CivicRAG has higher custom context relevance (0.9667 versus 0.9333), corresponding to 29 versus 28 positive context-set judgments. These results describe different aspects of behaviour and do not establish a single overall

Table 5.3: Completed saved-answer evaluation: Qwen3 self-judge scores, not independent accuracy.

Metric	Simple	CivicRAG	Δ	Pairs
Faithfulness	0.7983	0.7323	-0.0660	30
Answer relevancy	0.8460	0.8375	-0.0085	30
Custom context relevance	0.9333	0.9667	+0.0333	30

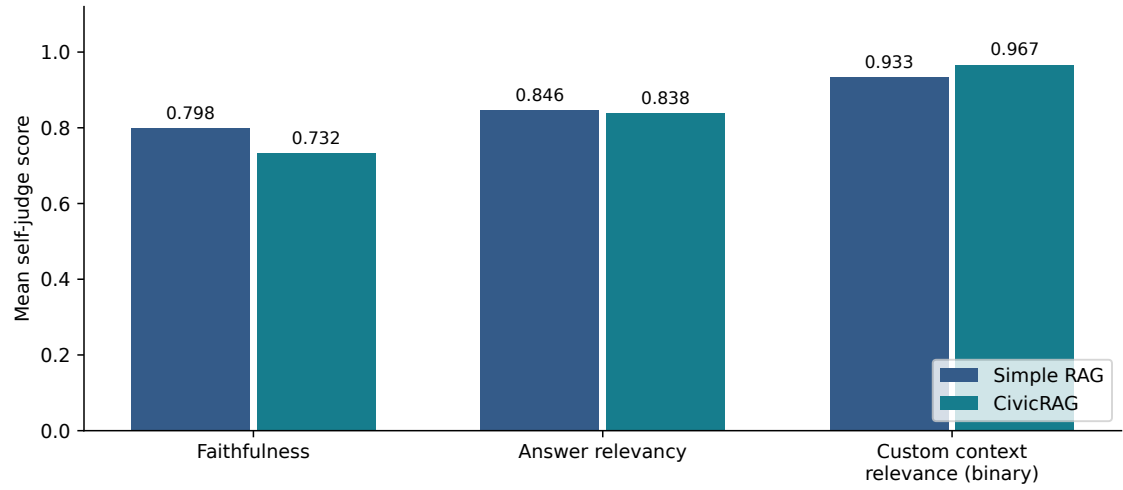


Figure 5.3: Completed local self-judge scores for matched saved answers, with 30 valid scores per system and measure. Custom context relevance is binary and distinct from standard context precision/recall.

winner.

Provisional Retrieval-Only Evaluation. A separate experiment compares BM25-only, dense-only and weighted Hybrid RRF on the same 30 questions before reranking, registration boosts and evidence selection. No answer generation occurs in this experiment. The top-five result union for each question forms a relevance pool; an assistant reviewed 309 question/passage pairs representing 192 unique passages. The labels are 105 relevant, 199 irrelevant and five uncertain.

These are unblinded, single-assistant labels, not independently human-verified ground truth or original-document legal verification. Four questions containing uncertain labels are excluded from all three methods, leaving 26 for Hit, Precision and MRR. One further question has no relevant passage in its judged pool; pooled Recall and nDCG therefore use 25 questions. This exclusion does not prove that the entire database lacks useful evidence.

The following equations specify the ranked-retrieval measures used in this study. Let z_{qi} indicate whether the chunk at rank i is labelled relevant, and let R_q be the relevant IDs in the judged pool. For each eligible question:

$$\text{Precision@5}(q) = \frac{1}{5} \sum_{i=1}^5 z_{qi}, \quad (5.2)$$

$$\text{Hit@k}(q) = \mathbf{1} \left[\sum_{i=1}^k z_{qi} > 0 \right], \quad k \in \{1, 5\}, \quad (5.3)$$

$$\text{RR@5}(q) = \begin{cases} 1/r_q, & r_q \leq 5, \\ 0, & \text{no relevant result in top five,} \end{cases} \quad (5.4)$$

$$\text{Recall@5}_{\text{pool}}(q) = \frac{|\text{Top5}(q) \cap R_q|}{|R_q|}. \quad (5.5)$$

MRR@5 averages RR@5 over eligible questions. Recall is undefined for an empty relevant pool. Following the normalized cumulative-gain framework [14], with binary relevance the discounted measure is

$$\text{nDCG@5}_{\text{pool}}(q) = \frac{\sum_{i=1}^5 z_{qi} / \log_2(i+1)}{\sum_{i=1}^{\min(5, |R_q|)} 1 / \log_2(i+1)}. \quad (5.6)$$

Pooled Recall and nDCG use only relevant passages found in the pooled rankings, not exhaustive corpus labels. Missing retrieved positions contribute zero to Precision’s fixed denominator of five. Near-duplicate IDs remain distinct scoring units.

Table 5.4 gives overall results. Asterisks denote pooled, rather than corpus-wide, relevance denominators. Dense retrieval leads on Hit@1, Precision@5, pooled Recall@5,

Table 5.4: Overall: provisional retrieval-only scores with assistant-reviewed labels. Hit, precision and MRR use 26 questions; pooled recall and nDCG use 25.

Method	H@1	H@5	P@5	R@5*	MRR@5	nDCG@5*
BM25	0.3077	0.6538	0.2769	0.3523	0.4179	0.3553
Dense	0.6923	0.8846	0.5000	0.7221	0.7821	0.7156
Hybrid RRF	0.4231	0.8846	0.4462	0.6173	0.6263	0.5863

MRR@5 and pooled nDCG@5. Dense and Hybrid RRF tie on overall Hit@5 (0.8846). Thus adding BM25 through these fixed fusion weights does not improve every first-stage ranking measure. These provisional results neither invalidate all uses of hybrid retrieval nor prove the complete answer pipeline inferior: later reranking and selection are deliberately absent.

5.2 Analysis of Design Solutions

What the Diagnostics Show. Generation-only trials supplied manually inspected passages for three known questions, then varied evidence order and added distracting passages. Better responses with relevant-only evidence still used the LLM; they were not controlled template answers. The trials showed why context selection deserved attention, but did not establish an automatic selector’s performance on arbitrary queries. This distinction is consistent with research separating retrieval quality from context use [10, 12].

The final matched comparison tests the automatically selected evidence rather than those manually curated diagnostic contexts. The RAGAS result is mixed: more positively judged context sets coexist with lower mean faithfulness. One possible explanation is that the remaining context contains conditions the generator combines incorrectly, but the current comparison does not isolate that mechanism. A context score cannot replace inspection of the answer.

Illustrative Outputs Provided by the Authors. The following three examples were supplied with the thesis revision request. All are reported with route `llm_selected_evidence`. Their reported times differ from the frozen comparison rows, whose corresponding elapsed times are 204.17, 147.21 and 180.81 seconds. We therefore retain the examples as author-supplied qualitative demonstrations and do not insert their timings or wording into the aggregate benchmark. The excerpts below preserve selected supplied sentences; they are neither gold answers nor current official instructions.

For passport services, the question is `পাসপোর্ট হারিয়ে গেলে কী করতে হবে?`, with a supplied time of 240.45 seconds. The response includes:

হারানো পাসপোর্টের ক্ষেত্রে অথবা চুরি হলে দ্রুত নিকটস্থ থানায় জিডি (GD) করতে হবে।

পুনরায় পাসপোর্টের জন্য আবেদনের সময় পূর্ববর্তী পাসপোর্টের ফটোকপি এবং জিডি কপি সহ আবেদনপত্র দাখিল করতে হবে।

The response addresses replacement rather than unrelated passport collection. Its additional identity-document and occupation-specific requirements still need condition-by-condition verification: a useful core instruction does not make every appended requirement applicable to every applicant. The route confirms model generation after selection, not independent validation.

For birth registration, the question is জন্মনিবন্ধন অনলাইনে কীভাবে করব?, with a supplied time of 176.82 seconds. The response directs the user to <https://bdris.gov.bd> and includes:

সমস্ত প্রয়োজনীয় তথ্য পূরণ করুন এবং আবেদন সাবমিট করুন।
দুইটি জমজ সন্তানের ক্ষেত্রে, প্রথম সন্তানের জন্ম নিবন্ধন শেষ হওয়ার পরে দ্বিতীয় সন্তানের জন্ম নিবন্ধন করতে হবে।

The portal-oriented answer aligns with an online application question. However, the unsolicited twin-specific condition shows that source-related content can enter a general answer without being requested. This example illustrates the need to separate general steps from exceptional cases, rather than assuming additional detail is always an improvement.

For BRTA, the question is BRTA-এর ড্রাইভিং পরীক্ষায় কী কী থাকে?, with a supplied time of 183.21 seconds. The response lists:

লিখিত পরীক্ষা
মৌখিক পরীক্ষা
ব্যবহারিক পরীক্ষা

The short list is more directly usable than a long legal extract. Nevertheless, the actual comparison trace discussed in Section 4.4 contains neighbouring licence topics, so this apparently clear answer is not evidence that all selected passages were applicable. Its supplied demonstration wording also differs slightly from the stored response. Across the three examples, local response latency remains substantial for interactive use.

Paired Answer Observations. Saved answers show both useful changes and regressions. Passport-document and fee answers become more task-aligned in CivicRAG, whereas the birth-registration timing response substitutes an application deadline for a processing time. Driving-licence status remains problematic. These examples explain

why the evaluation retains multiple measures rather than declaring correctness from answer length or source IDs.

The authors’ domain-informed preference for some more complete answers is useful qualitative feedback. Without a completed consistent review sheet, it is not converted into an aggregate expert accuracy percentage. Likewise, automatically assigned context labels cannot be relabelled as human judgments.

5.3 Final Design Adjustments

The evaluated configuration includes automatic applicability screening and the compatible-family ordering correction. Previously, compatible sibling fee sections could be appended after other candidates and then excluded by the context budget. Grouping them near the first eligible anchor changed the selected IDs for one question in a saved 30-case replay. This is a measured replay observation, not proof of general answer improvement.

The main CivicRAG answer file was generated after that correction. Its corpus, prompts and configuration were frozen for the matched comparison. No architecture change, model retraining or new answer generation was introduced while computing the reported metrics. The selected-evidence path bypasses older controlled-answer templates; the qualitative examples should not be described as those templates’ output.

Documentation refinements, including birth-registration terminology and the architecture diagrams, do not alter the underlying collection identifiers, code or saved results. The label retained in stored files is a provenance identifier, not an expansion of the evaluated service scope.

5.4 Statistical Analysis

Survey percentages use valid item responses rather than a fixed denominator for every question. Unknown recruitment, demographic concentration and unverified repeat participation prevent nationally representative inference. These descriptive needs-analysis statistics are separate from the paired system results.

For each automatic metric, define $\delta_q = s_{\text{Civic},q} - s_{\text{Simple},q}$. All 30 pairs have valid scores. The saved evaluator resamples the 30 paired differences with replacement 2,000 times, using seed 20260924, and reports empirical 95% intervals. Table 5.5 reproduces those saved intervals rather than recomputing a different protocol. The faithfulness and answer-relevancy intervals span zero; the custom binary context-relevance interval reaches zero.

Table 5.5: Paired CivicRAG minus Simple RAG differences with descriptive bootstrap intervals.

Metric	Mean Δ	95% interval
Faithfulness	-0.0660	[-0.1787, 0.0410]
Answer relevancy	-0.0085	[-0.0347, 0.0194]
Custom context relevance	+0.0333	[0.0000, 0.1000]

They reflect question resampling within this reused set, not judge calibration, repeated-generation variance or corpus representativeness. We therefore make no statistically established claim of overall factual superiority.

Per-domain retrieval results are shown in Tables 5.6–5.8. Values are question-level means over eligible questions; the overall score weights the eligible questions, not the three domain means equally. Dense and hybrid search both achieve passport Hit@5 of 1.0000

Table 5.6: Passport: provisional retrieval-only scores with assistant-reviewed labels. Hit, precision and MRR use 8 questions; pooled recall and nDCG use 8.

Method	H@1	H@5	P@5	R@5*	MRR@5	nDCG@5*
BM25	0.5000	0.7500	0.3000	0.3088	0.5833	0.3798
Dense	0.7500	1.0000	0.6500	0.7173	0.8750	0.7607
Hybrid RRF	0.3750	1.0000	0.4750	0.5253	0.6562	0.5328

Table 5.7: Birth registration: provisional retrieval-only scores with assistant-reviewed labels. Hit, precision and MRR use 9 questions; pooled recall and nDCG use 9.

Method	H@1	H@5	P@5	R@5*	MRR@5	nDCG@5*
BM25	0.3333	0.6667	0.2667	0.4370	0.4296	0.3938
Dense	0.5556	0.7778	0.4222	0.6963	0.6481	0.6173
Hybrid RRF	0.4444	0.8889	0.4222	0.6685	0.6148	0.5882

on eight eligible questions. For birth registration, hybrid Hit@5 is 0.8889 versus dense 0.7778 on nine questions, while dense retains stronger MRR. In BRTA, dense Hit@5 is 0.8889 versus hybrid 0.7778. These differences show why per-domain analysis is useful, but the small eligible samples and provisional labels limit generalization.

Uncertain questions excluded consistently across methods concern passport fees, passport renewal, birth-date proof and licence renewal. The licence-status pool contains no labelled relevant passage and is omitted only from pooled Recall/nDCG denominators. Uncertain labels are not treated as irrelevant, and missing scores are not replaced with zeros.

Exploratory Overnight Labeling. A separate Qwen3 labeling job completed 455 judgments over saved candidate and supplied passages: 176 relevant, 191 irrelevant and 88

Table 5.8: BRTA: provisional retrieval-only scores with assistant-reviewed labels. Hit, precision and MRR use 9 questions; pooled recall and nDCG use 8.

Method	H@1	H@5	P@5	R@5*	MRR@5	nDCG@5*
BM25	0.1111	0.5556	0.2667	0.3006	0.2593	0.2875
Dense	0.7778	0.8889	0.4444	0.7560	0.8333	0.7809
Hybrid RRF	0.4444	0.7778	0.4444	0.6518	0.6111	0.6377

uncertain. It hides system names, ranks and responses from the judge, but shares the generator’s model family. Under its stricter requirement for complete, nonempty relevant pools, only one of 30 questions is scorable. This is a completed labeling run with incomplete adjudication, not a completed 30-question retrieval benchmark.

The single scorable passport question has Precision@5 of 0.8 for Simple RAG and 1.0 for CivicRAG, with pooled Recall@5 of 0.8 and 1.0. Those numbers cannot support a system-level conclusion. This pool differs from the three-retriever top-five pool and its labels were not merged with the assistant-reviewed study. In particular, 455 judgments and 309 judgments are not two versions of one gold dataset.

5.5 Comparisons and Relationships

The secondary model runs use the same selected-evidence configuration and matching ordered context content. Table 5.9 summarizes operational outcomes; no equivalent RAGAS comparison is claimed for these secondary models. Qwen3 was a practical choice

Table 5.9: Secondary generator runs on the selected-evidence configuration. Language rejections are operational events, not factual error counts.

Model	Requests	Mean seconds	Language rejections
Qwen3:8b	30	181.82	0
Llama3	30	148.64	7
Qwen2.5:7b	30	147.38	1

for the main comparison, not an established best model for Bangla. Its model family supports the local deployment [4]; the observed outcomes belong to these particular recorded runs. Language rejection does not prove a response was factually wrong, and acceptance does not prove it was correct.

The comparison has two distinct levels. The answer-pipeline experiment holds generator settings fixed and varies retrieval plus later evidence processing. The retrieval-only experiment compares ranking before those later stages. A strong first-stage dense ranking can coexist with a helpful hybrid-pipeline answer on an individual question; conversely,

a positive context-set judgment can coexist with an unsupported generated statement.

The earlier 58-question registration-focused retrieval study uses different questions and labels and is not pooled into the current tables. NextRAG informs the method [9], but its financial-domain scores are not benchmark competitors here. No same-dataset experiment against that external system was performed.

5.6 Discussions

The strongest established contribution is a source-traceable dataset and a working integration that permits inspection from document preparation to selected evidence and answer. The comparison also identifies concrete strengths and limitations: CivicRAG improves some task-aligned outputs and the custom context-set score, while the dense baseline has stronger mean faithfulness and several first-stage retrieval measures.

The findings do not justify abandoning evidence inspection merely because an automatic mean is high. Both a passage’s applicability and the generator’s interpretation matter. The automated evaluation measures text support and relevance, whereas current legal validity, completeness for an applicant and practical usability require additional review.

The completed RAGAS calculation, provisional retrieval study and exploratory labeling run should therefore remain visibly distinct. The study demonstrates implemented capabilities and reproducible measurements without claiming universal superiority, 90–95% accuracy or readiness for unsupervised official advice. Further improvement can build on the recorded failure cases and source provenance rather than changing several components without a controlled comparison.

Chapter 6

Conclusion

6.1 Summary of Findings

This thesis developed Civic.ai as a locally executable RAG-based government-service information system focused on birth registration, passport and BRTA services. Its primary contribution is the preparation and organization of a source-traceable knowledge resource. The active collection contains 3,709 chunks from 174 prepared source units, with search-oriented text separated from evidence content. Combined registration documents remain in the underlying collection, but the study does not claim evaluated death-registration capability.

The implementation adapts established RAG methods [1, 9]: BGE-M3 dense retrieval, BM25, weighted RRF, reranking, automatic applicability selection and local generation. It preserves the evidence supplied to the generator and records response routes and configuration information. This makes both successful answers and failures inspectable without claiming that every source-linked response is correct.

Thirty Bangla questions were answered by CivicRAG and a matched dense-only Simple RAG baseline using Qwen3:8b. All 60 requests completed. The 180 saved-answer evaluation calculations also completed, giving mixed local self-judge results: Simple RAG has higher mean faithfulness and slightly higher answer relevancy, while CivicRAG has higher custom binary context relevance. These measures demonstrate observed behaviour, not independent factual accuracy or uniform superiority.

The separate provisional retrieval comparison provides further detail. Dense retrieval leads most overall measures, while Hybrid RRF ties dense Hit@5 overall and has higher birth-registration Hit@5 on the eligible subset. These assistant-reviewed results exclude uncertain questions and use pooled relevance, not exhaustive corpus ground truth. The overnight Qwen labeling run remains exploratory because only one question is scorable

under its current rules.

The needs-analysis survey supplies a different kind of evidence: 55 of 77 valid adult willingness responses (71.4%) favoured the proposed assistant, while 42 of 76 trust responses (55.3%) would recheck its information. This supports the design emphasis on understandable guidance and source inspection, without demonstrating adoption or population-wide social impact.

6.2 Contributions to the Field

The principal contribution is dataset development. We organized heterogeneous civic-service materials into domain-specific retrieval records, retained original-to-derivative relationships, and preserved available metadata, URLs and unresolved review flags. This resource supports systematic investigation of civic questions rather than relying on a general model’s untraceable knowledge.

The second contribution is the developed system integration. Search aliases support retrieval without becoming answer facts; chunk identifiers connect vector results to source content; and selection traces make post-retrieval decisions visible. The architecture adapts established techniques to the language and procedure distinctions of the collected documents rather than claiming a new foundation model or general-purpose RAG algorithm.

The third contribution is an auditable evaluation record. Matched baseline answers, selected contexts, model settings, local judge scores and provisional retrieval labels are retained separately. Manually selected diagnostics are not confused with automatic operation, and completed computations are not confused with validated ground truth. This separation allows future work to test specific changes without losing the current comparison.

6.3 Recommendations for Future Work

The first priority is independent, evidence-backed annotation of unseen questions and relevant passages. The set should cover paraphrases, conditional scenarios, ambiguous questions and insufficient-evidence cases across the three studied domains. Additional services, including death registration, would require explicit scope definition and evaluation rather than assuming that their appearance in a combined source proves support.

Data review should address unresolved original matches, outdated instructions, conflicting conditions and long legal OCR. Reviewed changes should be versioned with the corpus and rebuilt indexes. Relevance annotation and verification against current official docu-

ments serve different purposes and should be recorded separately.

Architectural improvements should use single-factor comparisons. Selector-on versus selector-off, adjusted context budgets and dense retrieval followed by reranking are useful candidates. Stronger multilingual generators or independent evaluation models may improve particular stages, but their benefits should be measured rather than inferred from model size or a persuasive demonstration. More capable hardware and measured runtime profiling could address the recorded interactive latency.

Finally, a prospective user study could measure clarity, usefulness and the effort needed to verify responses. Such evidence would be needed to substantiate the intended social outcomes described in the preserved Abstract. Civic.ai is presented as a developed and evaluated information system with a documented dataset contribution and bounded findings, not an official decision-making service.

Bibliography

- [1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474. [Online]. Available: <https://arxiv.org/abs/2005.11401>
- [2] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “BGE M3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” arXiv:2402.03216, 2024. [Online]. Available: <https://arxiv.org/abs/2402.03216>
- [3] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, “Reciprocal rank fusion outperforms condorcet and individual rank learning methods,” in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2009, pp. 758–759.
- [4] A. Yang, A. Li, B. Yang *et al.*, “Qwen3 technical report,” arXiv:2505.09388, 2025. [Online]. Available: <https://arxiv.org/abs/2505.09388>
- [5] C. Auer, M. Lysak, A. Nassar, M. Dolfi, N. Livathinos *et al.*, “Docling technical report,” arXiv:2408.09869, 2024. [Online]. Available: <https://arxiv.org/abs/2408.09869>
- [6] S. Robertson and H. Zaragoza, “The probabilistic relevance framework: BM25 and beyond,” *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.
- [7] R. Nogueira and K. Cho, “Passage re-ranking with BERT,” arXiv:1901.04085, 2019, preprint. [Online]. Available: <https://arxiv.org/abs/1901.04085>
- [8] S. Es, J. James, L. E. Anke, and S. Schockaert, “RAGAs: Automated evaluation of retrieval augmented generation,” in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System*

Demonstrations. Association for Computational Linguistics, 2024, pp. 150–158. [Online]. Available: <https://aclanthology.org/2024.eacl-demo.16/>

- [9] S. A. I. Opin, R. A. Thakur, A. K. Raiyan, M. R. Kabir, and S. Rahman, “NextRAG: A bilingual retrieval-augmented generation framework for financial question-answering,” in *2025 28th International Conference on Computer and Information Technology (ICCIT)*. IEEE, 2025. [Online]. Available: <https://doi.org/10.1109/ICCIT68739.2025.11491522>
- [10] A. S. Ipa, M. A. T. Rony, and M. S. Islam, “Empowering low-resource languages: TraSe architecture for enhanced retrieval-augmented generation in bangla,” in *Proceedings of the 1st Workshop on Language Models for Underserved Communities (LM4UC 2025)*. Association for Computational Linguistics, 2025, pp. 8–15. [Online]. Available: <https://aclanthology.org/2025.lm4uc-1.2/>
- [11] R. Aich, N. Das, and M. Chowdhury, “HybridRAG-BN: A retrieval-augmented framework with fine-tuned verification for bangla KBQA,” arXiv:2608.13004, 2026, preprint. [Online]. Available: <https://arxiv.org/abs/2608.13004>
- [12] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, “Lost in the middle: How language models use long contexts,” arXiv:2307.03172, 2023. [Online]. Available: <https://arxiv.org/abs/2307.03172>
- [13] Qwen Team, “Qwen2.5 technical report,” arXiv:2412.15115, 2024, first submitted December 2024; subsequent revisions available. [Online]. Available: <https://arxiv.org/abs/2412.15115>
- [14] K. Järvelin and J. Kekäläinen, “Cumulated gain-based evaluation of IR techniques,” *ACM Transactions on Information Systems*, vol. 20, no. 4, pp. 422–446, 2002.

Appendix A

How to install L^AT_EX

This appendix retains the title in the official template and describes the build requirements for this manuscript. The source uses XeTeX-compatible Unicode fonts. The verified local build uses Tectonic, which handles compilation passes and bibliography processing.

1. Install a compatible TeX toolchain. This manuscript was verified with Tectonic; Xe-LaTeX with BibTeX is an alternative.
2. Ensure Times New Roman and Kohinoor Bangla are available. If replacement fonts are necessary, change only the font declarations in `main.tex` and recheck all pages. The Abstract text must remain unchanged.
3. Keep `core`, `chapters`, `appendix`, `bibliography` and `generated` beside `main.tex`.
4. From the manuscript directory run `tectonic --keep-logs --keep-intermediates main.tex`.
5. Review the PDF, contents pages, equations and tables. Resolve missing characters, undefined references and overflowing text before submission.

The generated tables are included in the source package. Rebuilding the PDF does not require running the chatbot or downloading language models. Updating experimental results is a separate operation: the repository audit script reads saved artifacts and checks their configuration hashes before regenerating tables.

Appendix B

Overleaf: GitHub for L^AT_EX projects

The original template’s appendix title is retained. The manuscript is stored with its L^AT_EX source, bibliography and generated tables in the project repository. Git provides revision history and rollback points; the PDF is a compiled output rather than a substitute for editable source.

For hosted editing, transfer the source package while preserving its directories, select `main.tex` as the main document and use a Unicode-compatible compiler. Local fonts may not be available in the hosted environment. Use legally available fonts and review any layout changes. This manuscript has been compiled locally; a successful hosted build is not claimed.

Before revising the report, preserve the current Git revision. Keep manuscript edits separate from evaluation artifacts so that a reporting change does not silently alter an experiment. Compile and review the PDF before committing its source and output together. Keep incomplete scores distinct from completed results and never overwrite the unchanged Abstract while merging revisions.

Additional reproducibility notes, evaluation questions and completion requirements remain in the accompanying source package and repository. They are supporting artifacts, not replacement appendix titles or added entries in the official template’s chapter index.